

Prompt engineer notes by durga sir Day wise

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Day:1

Give me strong ginger tea.

Less sugar.

In a small cup.

What is prompt?

It is an instruction given to AI.

clear prompt gives clear output.

Write a birthday message for my friend.

What to do ---> write message

For whom ---> my friend

For what --> birthday

AI understands

AI writes

Bad vs Good Prompts:

What is a bad prompt?

unclear

too open

missing details

AI must guess

eg: Tell about job?

AI provides very generic answer.

What is a good prompt?

clear

specific

tells audience

tells style

Eg: Explain about IT jobs for freshers in simple english.

Topic = IT Jobs

Audience = Freshers

Language = Simple english

Bad prompt creates confusion

Good prompt creates direction

If prompt is unclear

AI gives general answer

else

AI gives specific answer.

Eg-1: Explain python?

which level

which part

short or long

example needed or not

Explain python for a 10th class student.

Use very simple words.

Give one daily life example.

Audience is clear

language is clear

output style is clear

example is requested

Eg2:Write an email?

to whom?

purpose?

tone?

Write a professional email to my manager.

Topic: Request leave for 2 days.

Reason: feverkeep it polite and short.

Eg: Interview preparation

Give interview questions

which subject?

which level?

how many?

Ask me 5 prompt engineering interview questions for fresher.

Wait for my answer after each question.

How to write good prompt?

A good prompt should be specific and clear.

It tells what I want and for whom.

Note:

Bad prompt gives general answer.

Good prompt gives useful answer.

prompt clarity rules:

Rule-1: Specific vs Vague

Vague means:

not clear

too general

missing details

Eg: explain java

Specific prompt:

Clear

focused

exact need mentioned

Explain java basics for freshers in simple indian english with one example.

Topic java

Audience Freshers

Language = simple english

style with example

Rule-2: Complete vs Incomplete:

incomplete means

half information

missing information

missing goal

eg: write resume

complete prompt means:

All needed information is given

write resume for python fresher with no experience.

Focus on skills and projects.

use simple professional language.

If prompt is vague or incomplete(bad):

AI guesses

output becomes weak

else

AI understands clearly

output becomes strong

Day2:

Prompt Formula:

RTC CO

Role + Task + Context + Constraints + Output

1. Role: Who should AI act like?

What role should AI play?

eg:

Act as a Teacher

Act as a HR

Act as an interviewer

Act as a manager

Act as a mentor

Role gives direction to the AI.

Without role, AI gives general answer.

2. Task - What should AI do?

what work you want AI to do

eg:

Explain Python

Write email

Create resume

Ask interview question

Task gives job to AI.

3. Context - For whom/background:

Background information

eg:

For school student

For fresher

For working professional

For business owner

Context tells AI level and situation.

4. Constraints- Rules and Limits:

means restrictions or conditions

eg:

use simple english

do not use technical words

Answer in 5 points

keep it short

constraints control AI behaviour

5. Output Format - How answer should look like:

How do you want the answer?

eg:

Bullet points

Table

steps

short paragraph

Format gives shape to the output.

Set Role

Set Task

Set Context

Set Constraints

Set Output format

eg1: Student Learning

Act as a school teacher

Explain python loops

Audience: beginner student

Use Simple Indian English

Give one daily life example

g2: Interview Practice

Act as an interview person

ask me 5 Exadata interview questions

Level: 10 years Experience

wait for my answer after each question

Act as Exam Expert

provide me 3 MCQs on prompt engineering

Level: fresher

Ask questions one by one

Wait for my answer for each question

Please validate my answer

provide correct answer with explanation with 3 bullet points

4. Official Email:

Act as a professional assistant

write email to manager

context: requesting 2 days leave

Tone: polite

keep it short

please provide email to my manager about 2 days leave

5. Resume Building:

Act as HR.

Create a resume summary

context: Python fresher

Focus on skills and projects

use professional language

6. Coding Help:

Act as programming mentor

Explain if-else in python

Audience: beginner

Explain step by step

7. Business Marketing:

Act as Marketing Expert

Write instagram caption

context: small mobile shop

Tone: friendly

Add call to action

Day03:

Just remember 5 buzz words:

Role Task Context Constraints Output

Role gives direction

Task gives work

Context gives clarity

Constraints give control

Output gives shape

8. Office Task planning:

Act as project manager

create daily task list

context: Software developer

Format: bullet points

9. Study Time table:Act as study mentor

create study time table

context: college student

constraints: 2 hours per day

Act as Company Owner

Create action plan to start an AI based software company starts from scratch

context: Zero technical knowledge

First month he can spend only 5 mins per day, add 5 more minutes for every next month

Provide action plan week by week.

For every week wait for my confirmation

Act as Experienced AI Trainer

Provide 30-days action plan to learn Generative AI from basic to expert level

day by day. After every day wait for my confirmation

context: Beginner

constraints: per day time availability is just 30 mins

Output: table format

10. Explanation Control:

Act as teacher

Explain SQL Joins

Audience: non-technical people

Do not use technical terms

Act as Experienced Lover Boy:

Explain top 3 ways to impress a girl

context: fresher

constraints: without spending much time and money

output: bullet format

Notes Generator:

Act as exam coach

Summarize prompt engineering

Audience: beginner

output: 5 bullet points

 Prompt Engineering – 5 Key Points

Prompt engineering means giving clear instructions to AI to get correct and useful output

A good prompt explains what to do, how to do, and expected result

Clear prompts reduce wrong answers and confusion from AI

Changing words in a prompt can change the output completely

Prompt engineering helps in writing content, code, emails, answers, and ideas faster

Q. How do you write good prompt?

We use Role, Task, Context, Constraints and Output format

Give one example

Tone Control:

Tone means:

style of speaking

mood of speaking

how the answer should "feel".

4 common tones we use:

1. Friendly
2. Professional
3. funny
4. strict

1. Friendly:

means

soft

simple

caring

like a helpful friend

where we can use:

beginner

school students

people who fear technical topics

2. Professional:

means:

office style

formal

clean

polite

no jokes

usage:

emails

reports

client messages

linkedin posts

3. Funny:

means:

jokes

light style

fun examples

makes boring topic interesting

usage:

reels

youtube shorts

marketing content

4. strict:

means:

direct

serious

no extra words

like strict teacher or coach

usage:

interview preparation

quick revision

strong discipline style learning

if tone = friendly

answer becomes soft and simple

if tone = profesional

answer becomes formal and office style

if tone = funny

answer includes jokes and fun examples

if tone = strict

answer becomes direct and serious

friendly:

Explain prompt engineering in friendly way for a beginner

use simple english

give one daily life example

Explain prompt engineering in professional way

write like an office trainer

use clean bullet points

Explain prompt engineering in funny way

use one chai shop example

make it entertaining.

Explain prompt engineering strictly for interview

No jokes

Answer in 5 bullet points only

Q. How to control tone in AI output?

By clearly mentioning tone in the prompt like friendly or professional and also adding constraints like no jokes or formal language.

Day04:

Length Control:

we can tell AI how big the answer should be.

You decide:

short

medium

detailed/long

1. short:

used for :

quick revision

interviews

summaries

Explain Java in 2 lines.

2. medium:

used for:

learning

notes

normal explanation

Explain Java in 200 words.

3. long/detailed:

used for:

deep understanding

teaching

documentation

Explain Java in detail with examples.

Note:

If we don't mention length,

AI will guess.

Guessing can give:

too long

or too short

which may not fulfill your requirement.

Always control length.

If length is defined

AI controls answer size

Else

AI gives random size

Explain Python in around 100 words

audience: beginner

Give one daily life example.

step-by-step prompting:

You tell AI to explain in ordered steps.

This avoids confusion.

eg: Explain SQL joins step by step.

step 1: What is join

step 2: Why join is required

step 3: simple example

eg: step-by-step + length control

Explain for loop in python

audience: beginner

Explain step by step

each step in 2 lines only.

eg: Office Task

Create daily work plan step by step

Audience: software employee

keep each step one line only

eg: durga sir prompt engineering class plan

Create daily prompt engineering class of 5 mins, plan step by step

Audience: AI victims

keep each step 2 lines only

If step-by-step is mentioned

AI explains in order

else

AI mixes explanation

Q. How do you avoid long AI Answers?

By clearly mentioning length and using step-by-step prompts

If I practice daily

confidence increases

If confidence increases

fear reduces

If fear reduces

usage increases

If usage increases

mastery happens and happily you can settle your life in this AI era

Day05:

Mastering AI Format control: Messy text to professional output

Format Control = Telling AI how the answer should be presented.

Without format control, AI answer looks like raw text.

With format control, AI answer looks professional.

Format 1: Bullet Points

short

clean

easy to read

eg:

Explain prompt engineering advantages in 5 bullet points

every point in a single line

Helps AI understand exactly what we want and give correct output

Saves time by reducing repeated prompts and corrections

Allows control over tone, length, and format of the response

Improves quality of answers for coding, logic, and problem solving

Makes AI more useful for real-world work and daily tasks

usage:

Notes

Interviews

Presentations

Format-2: Table

comparison

clarity and easy to understand

structured view

eg:

Explain Python vs Java in a table with 3 columns

(Language | Use | Difficulty)

Usage:

comparison

reports

clarity

Format 3: Steps:

Ordered explanation

no confusion

eg:

Explain how to prepare for prompt engineer interview in step by step format

Here is a step-by-step plan to prepare for a Prompt Engineer interview (simple Indian English) 👉

Step 1: Understand the basics clearly

What is a prompt, why prompt quality matters, and how AI responds to instructions

Learn: role, task, context, constraints, output format

Step 2: Practice writing prompts daily (30–60 mins)

Take 1 normal question and write 5 prompt versions:

simple, professional, strict, funny, and detailed

Step 3: Learn common prompt types

Zero-shot (no example)

One-shot (1 example)

Few-shot (2–5 examples)

Learn when to use which

Step 4: Learn “reasoning prompts” safely

For logic tasks, use structured prompting like:

“Explain the steps clearly”

“Show final answer + short reasoning”

Practice 10 logic questions using structured prompts

Step 5: Learn prompt debugging (very important)

If output is wrong, fix the prompt by:

adding missing contextgiving constraints (time, budget, audience)

asking for specific format (table / bullets / JSON)

asking AI to ask questions before answering

Step 6: Build a small Prompt Library

Create ready templates for:

Email writing

Resume + LinkedIn

Summarization

Content creation (Instagram/YouTube)

Coding help (bug fix + explain)

Data analysis tasks

Step 7: Learn evaluation mindset (interviewer likes this)

When you get an output, check:

correctness

completeness

clarity

safety (no wrong advice)

consistency in format

Step 8: Prepare 5–6 strong project stories (must)

Example projects you can say:

Built prompts for customer support replies

Created prompt templates for marketing captions

Built Q&A bot using documents (RAG concept)

Automated report generation using prompts

Prompt testing and improving accuracy

Step 9: Prepare for common interview questions

Practice answering:

“What is a good prompt?”

“Zero-shot vs Few-shot?”

“How do you reduce hallucinations?”

“How do you control output format?”

“How do you test and improve prompts?”

Step 10: Do a mock interview practice

Take 10 prompts, time yourself, and explain:

why you wrote it

what output you expect

how you will improve if output fails

Used for:

learning

process explanation

planning

Format 4: Checklist

yes/no style

to-do list/priority list

completion focus

eg:

Create prompt engineer interview preparation checklist for fresher.

Used for:

daily tasks

tracking work

discipline

Format 5: story:

story format means

flow

emotion

easy understanding

eg:

Explain importance of savings using a short story

Used for:

motivation

reels

teaching beginners

Explain importance of exercise in 3 bullet points

Explain importance of exercise in a table with 2 columns

(Benefit | effect)

Explain importance of exercise in step by step

Create checklist for exercise activity

Explain importance of exercise with a story.

Q. How do you control AI output format?

Ans: By clearly mentioning the required format like bullet points, table, or steps in the prompt.

Home work:

"Benefits of learning AI"

Ask AI to give:

bullet points

table

checklist

Notes:

1. Format control decides how answer looks like.
2. Same content can be shown in many formats
3. Clear format instruction gives clean output.

Mastering Structured Output: Turning AI answers into professional Systems:

Structured output = Telling AI to always follow the same structure.

structure means:

fixed headings

fixed order

fixed table columns

eg With fixed headings:

Explain SQL using fixed headings:

1. What is it
2. Why it is needed
3. Simple example
4. Interview point

Explain prompt engineering using fixed headings:

1. What is it
2. Why it is needed
3. Simple example
4. Interview point

Where fixed headings are used?

Notes generation

Teaching material

PDFs

Reels scripts

Reusable prompts

eg2: strict table columns:

strict table means:

column names are fixed

order fixed

no extra columns

Explain Python data types in a table

columns must be exactly:

| Data Type | Example | Use |

Usage for:

Excel

Google sheets

Reports

Automation

structured vs unstructured :

unstructured:

Explain python list

With fixed headings:

Explain python list using fixed headings:

Definition

Key features

Simple example

Interview tip

Explain Python List using table with columns:

| Feature | Description |

Q. Why structure output is important in AI prompt?

Structured output makes AI responses predictable, reusable and professional, especially for reports and automation.

Notes:

Structured output means fixed format every time

Fixed headings bring clarity and consistency

strict tables make output reusable and professional.

DAY06:

"Only output this" control

By default AI talks extra.

We should stop it. For this we required to use "Only output this"

It telling to AI:

give only required format

do not add explanation

do not add greeting

do not summary

useful controls:

Reply only with output

do not add explanation

no greetings

no summary

eg:

Explain benefits of learning AI in 5 bullet points

Explain benefits of learning AI.

output format : 5 bullet points only

Do not add extra text

Explain benefits of learning AI

output format: table

columns must be exactly:

| Benefit | Explanation |

Do not add extra text

TOON

Convert benefits of learning AI into JSON.

Reply only in valid JSON.

Do not add any extra text.

Give interview preparation steps in bullet points only.

Do not add any extra text.

Give interview preparation steps in table with columns: | step | action |

Do not add any extra text.

Convert interview preparation steps into JSON.

Reply only in JSON.

Do not add any extra text.

Notes:

"Only output" control removes extra AI talk.

Q. Why only output control is important?

Because tools and automation need clean output without extra text.

Extra explanations break integrations.

Notes Generator Prompt:

Reusable prompt

Real world skill

prompt template

Step1: Define Role

Act as an experienced teacher

step2: Define task

Generate student notes for the given topic:{TOPIC}

STEP3: Defines structure(Fixed Headings)

1. Definition

2. Key points

3. Simple example

4. Interview Tip

step4: Define format rules:

Bullet points only

short and clear lines

step5: output control lines

Do not add introduction

Do not add conclusion

Do not add extra explanation

Act as an experienced teacher

Generate student notes for the given topic: {TOPIC}

Follow this fixed structure strictly:

1. Definition
2. Key points
3. Simple example
4. Interview Tip

Rules:

Use simple indian english

Do not add introduction

Do not add conclusion

Do not add extra explanation

output format:

use Bullet points only

keep each point short

Act as an experienced teacher

Generate student notes for the given topic: {TOPIC}

context: This notes required for non-technical students

Follow this fixed structure strictly:

1. Definition
2. Key points
3. Simple example
4. Interview Tip

Rules:

Use simple indian english

Do not add introduction

Do not add conclusion

Do not add extra explanation

output format:

use Bullet points only

keep each point short

Q. What is reusable prompt?

A reusable prompt is a fixed instruction template that gives consistent output for different inputs.

Day07:

Encoding Instructions inside prompt:

Conclusion-1: AI follows written instructions, not your intention.

AI cannot read your mind.

AI follows what you write.

Missing instruction = AI guesses

eg:

Act as beginner-friendly teacher

Explain python loops

in simple english.

Conclusion-2: Instruction order decides output quality

Early instructions set direction.

Role + Task should come first.

Conflicting instructions confuse AI.

eg:

Act as Interview Coach.

Explain SQL joins

output in 5 bullet points only.

Conclusion-3: Missing details create wrong output.

Missing audience = generic answer.

Missing format = unwanted paragraph

Missing control = too many outputs

eg:

Act as software trainer

Explain java for 10th class student.

Give exactly 3 bullet points.

Do not add extra text.

Act as software trainer

Explain java

Give exactly 3 bullet points.

Do not add extra text.

Why small words change big output:

Bring vegetables

Bring only onions

Bring only onions, do not bring anything else.

Bring exactly 1 kg onions.

Conclusion-1: Small control words change big output

"only" restricts output.

"exactly" fixes count.

"do not" stops extra content.

eg:

Explain AI.

Give only bullet points.

Conclusion-2: Precision prompting gives control

Clear rules reduce guessing.

Output becomes consistent.

Less confusion for AI.

eg:

Act as AI Trainer.

Explain prompt engineering for beginners.

Give exactly 5 points.

Do not add explanation.

Conclusion-3: Casual prompts create random answers

"Explain this" is unclear

AI takes freedom and varies output

precision = engineering mindset

casual = just chatting

eg:

Act as software trainer

Explain SQL for beginners.

Use simple english

output must be a table. **Ambiguity Problems - Why LLM guesses:**

Conclusion-1: Ambiguity means unclear instruction

Multiple meanings possible

Missing details create confusion.

AI must reply, so it guesses

eg:

explain cloud. (Ambiguous)

Conclusion-2: Clear prompts remove randomness

Add audience

Add purpose

Add format + constraints

eg:

Act as cloud expert:

Explain cloud computing for non-technical students.

Use simple english.

Give 1 daily life example.

Act as my girl friend.

please provide some jokes as I am feeling bore.

use simple english

short jokes only

Conclusion-3: Under specified prompts cause mistakes.

Missing level, format, goal

AI chooses its own assumptions

output may not match your requirement.

eg:

Act as experienced technical person

create a resume for python fresher.

Use 5 bullet points

Add 2 projects

Day08:

Zero-shot, One-shot and few shot prompts:

Zero-shot prompting:

Direct Asking

No examples

This is the most basic and most used prompting style.

eg: Explain Python

When you use zero-shot:

AI looks at your words

AI uses past training data

AI predicts best possible answer.

When zero-shot prompt is enough:

1. You want quick help
2. You are exploring topic
3. Format does not matter

zero-shot = speed + simplicity

Limitations of zero-shot prompting:

1. output variation
2. No style control
3. Not reusable

Note:

Zero-shot is good for learning not good for production.

eg:

Explain Java

What is OOPS?

Q. What is zero-shot prompting?

Zero-shot prompting means asking AI directly without giving any example.

Output may vary because AI guesses style.

One-shot prompting:

You give one example

Then ask AI to do same task

AI follows same pattern and style

One example = direction to AI

Note: From provided example AI understands expected format, tone and structure.

eg:

Example

Q. What is Java?

Ans: Java is Highlevel object oriented programming language.

Now explain python in same way.

Example

Q. What is Java?

1. Java is Highlevel object oriented programming language.

2. Java is platform independent

3. Java can be used to develop stand alone, web and enterprise applications

Now explain python in same way.

Example

Q1. What is big speciality of Java?

A) Platform dependent

B) Platform independent

C) Procedure oriented

D) functional oriented

Ans: B

Explanation: Java Follows Write once Run anywhere.

Now in the same style please generate 5 questions on zero-shot, one-shot and few-shot prompts

Example – MCQs on Zero-Shot, One-Shot, Few-Shot Prompts

Q1. What is Zero-Shot Prompt?

- A) Prompt with many examples
- B) Prompt with one example
- C) Prompt with no example
- D) Prompt with code only

Ans: C

Explanation: In Zero-Shot, AI is asked to do a task without giving any example.

Q2. What is One-Shot Prompt?

- A) Prompt without instructions
- B) Prompt with one example
- C) Prompt with many examples
- D) Prompt with system role only

Ans: B

Explanation: In One-Shot, one example is given to guide the AI.

Q3. What is Few-Shot Prompt?

- A) Prompt with zero examples
- B) Prompt with one example
- C) Prompt with 2–5 examples
- D) Prompt without task

Ans: C

Explanation: Few-Shot uses multiple examples so AI understands the pattern clearly.

Q4. Which prompt type is best for simple and clear tasks?

- A) Few-Shot
- B) One-Shot
- C) Zero-Shot
- D) Multi-Shot

Ans: C

Explanation: Zero-Shot works well when the task is simple and direct.

Q5. Which prompt type gives best accuracy for complex tasks?

- A) Zero-Shot
- B) One-Shot
- C) Few-Shot
- D) Random Prompt

Ans: C

Explanation: Few-Shot gives better accuracy because more examples reduce confusion.

Example:

"Computer is like a helper boy in shop - ask once, it remembers and brings"

In the same way, explain about Artificial intelligence

Example:

Internet is like gossip aunties network - news spreads fast.

In the same way explain about prompt engineering.

When to use One-shot prompt:

You want same pattern again and again

eg:

notes

quizzes

emails

captions

where ever consistency is required, one-shot is best

Q. What is one-shot prompting?

One-shot prompting means giving one example so AI follows the same style and structure.

It gives consistent output.

Zero-shot ==> Asking AI without any example

One-shot ==> Guiding AI with one example

Few-Shot ==> AI understands deeply with our provided multiple examples.

Day09:

Few-Shot prompting:

Multiple examples mostly 2 to 5

context learning

Better accuracy

Few shot prompting means:

You give 2 to 5 examples

AI learns the pattern

AI applies the same logic to new input

Why 2-5 examples work best:

Why not 1 example?

AI may just copy style.

Understanding is shallow.

Why not too many?

prompt becomes long

confusion increases

Tokens increases

2-5 examples = sweet spot

eg1: Customer support Reply:

Example 1:

Customer: My order is delayed.

Reply: Sorry for the delay. We are checking and will update you soon.

Example 2:

Customer: I received a damaged product.

Reply: Sorry about the issue. Please share photos. We will replace it.

Now reply to this:

Customer: I want to cancel my order.

AI answer:

Reply: Sorry to hear that. Please share your order details. We will help you with the cancellation process.

eg2: Product Description (E-commerce):

Example 1:

Product: Water Bottle

Description: Leak-proof bottle, easy to carry, good for daily use.

Example 2:

Product: Laptop Bag

Description: Strong bag with multiple pockets, suitable for office. Now write Description for

Product: Wireless Mouse

Product: Wireless Mouse

Description: Smooth and fast mouse, easy to connect, comfortable to use for office and daily work.

Example 1:

Product: Water Bottle

Description: Leak-proof bottle, easy to carry, good for daily use.

Example 2:

Product: Laptop Bag

Description: Strong bag with multiple pockets, suitable for office.

Now write Description for

Product: Eggs

Product: Eggs

Description: Fresh and healthy eggs, rich in protein, good for daily breakfast and cooking. Example 1:

Product: Water Bottle

Description: Leak-proof bottle, easy to carry, good for daily use.

Example 2:

Product: Laptop Bag

Description: Strong bag with multiple pockets, suitable for office.

Now write Description for

Product: Mobile

Product: Mobile

Description: Smart and easy-to-use mobile, fast performance, clear display, suitable for daily calls, browsing, and entertainment.

eg3: Whatsapp message writing:

Example 1:

Message : I will join office late today due to traffic.

Example 2:

Message : Please find the report attached. Let me know if changes needed.

Now write a short professional WhatsApp message for

Taking leave tomorrow due to fever.

Message: I will be on leave tomorrow due to fever. I will respond to urgent matters once I am available.

eg4: Few-Shot classifier:

Example 1:

"I love this phone" --> Positive

Example 2:

"This service is terrible" --> Negative

Example 3:

"The food was Okay" --> Neutral

Now classify:

"The movie is amazing"

Example 1:

"I love this phone" --> Positive

Example 2:

"This service is terrible" --> Negative

Example 3:

"The food was Okay" --> Neutral

Now classify:

"The match was boring"

Example 1:

"I love this phone" --> Positive

Example 2:

"This service is terrible" --> Negative

Example 3:

"The food was Okay" --> Neutral

Now classify:

"The AI class is average"

Few-Shot Prompts with Structured Output:

It is most commonly used in production. ie Production-Ready Prompting.

eg1: Invoice Extractor:

Finance team uploads invoices.

They need:

Invoice number

Date

Total Amount

But provided invoices have different formats.

Extract data in this format:

Invoice Number:

Date:

Total Amount:

Example 1:

Input:

Invoice No: INV-101

Date:01-02-2025

Total: Rs 15,000

Output:

Invoice Number: INV-101

Date:01-02-2025

Total Amount: Rs 15,000

Example 2:

Input:

Bill ID: 2045

Date:10 March 2025

Grand Total: Rs 22,500

Ouput:

Invoice Number: 2045

Date:10-03-2025

Total Amount: Rs 22,500

Now extract from this invoice:

Input:

Invoice #8899

Dated: 5 April 2025

Total Payable: Rs 40,000

Output:

Invoice Number: 8899

Date: 05-04-2025

Total Amount: Rs 40,000Extract data in this format:

Invoice Number:

Date:

Total Amount:

Example 1:

Input:

Invoice No: INV-101

Date:01-02-2025

Total: Rs 15,000

Ouput:

Invoice Number: INV-101

Date:2025-02-01

Total Amount: Rs 15,000

Example 2:

Input:

Bill ID: 2045

Date:10 March 2025

Grand Total: Rs 22,500

Ouput:

Invoice Number: 2045

Date:2025-03-10

Total Amount: Rs 22,500

Now extract from this invoice:

Input:

Invoice #8899

Dated: 5 April 2025

Total Payable: \$ 40,000

Day10:

Why Industry uses this?

Automated accounting

Consistent extraction

No extra text

Ready for ERP systems

Few-Shot with structured output:

Report Generator:

Company needs weekly sales report in fixed format.

Prompt:

You are a business report generator

Use this format:

Title:

Summary:

Key Points:

Conclusion:

Example 1:

Input:

Sales increased by 20%. Marketing campaign performed well.

Output:

Title: Weekly Sales Report

Summary: Sales increased significantly

Key Points:

-20% growth

-Marketing campaign success

Conclusion: Positive growth trend.

Example 2:

Input:

Customer complaints reduced by 30

Output:

Title: Weekly Performance Report

Summary: Customer satisfaction improved

Key Points:

-30% complaint reduction

Conclusion: Service quality improved.

Now generate report for this data:

Input:

Revenue increased by 15%. New clients added 50.

Title: Business Growth Report

Summary: Revenue and customer base have grown positively

Key Points:

Revenue increased by 15%

50 new clients added

Conclusion: Business is growing steadily with higher income and more customers.

Why Industry uses this:

Board-ready format

Clean structured output

predictable layout

Persona Prompting:

Role:

Role means job

eg:

Teacher

HR

Interviewer

Analyst

Role answers===>What is the job

Persona:

Persona means behaviour style

eg:

Teacher role can be:

Friendly teacher

Strict teacher

Motivational teacher

Funny teacher

Persona answers ===> How does this role behaves

Tone:

Tone means voice feeling

eg:

polite

serious

encouraging

Tone is smaller layer.

Persona is bigger behaviour identity.

Role= Job

Persona = Behaviour style

Tone = Voice feeling

Q. What is the difference between Role and Persona?

Role defines responsibility/Job

Persona defines behaviour identity of the role.

No Role:

Explain Python in simple english.

With Role :

Act as a teacher

Explain Python in simple english

With persona:

Act as a friendly school teacher

Explain python in simple english

Use 1 daily life example

end with 1 motivational line.

Act as a friendly technical trainer like durga sir

Explain python in simple english

Act as a strict interview coach

Ask me 1 question about prompt engineering

Wait for my answer

then give feedback in 3 bullet points

Day11:

Persona control(Behaviour Engineering):

In real industry:

HR bots must be polite

Financial bots must be strict and logical

Mentor bots must be supportive and motivational

Persona = Behaviour + Depth + Attitude

Behaviour:

Is AI strict?

Is AI friendly?

Is AI Funny?

Depth:

Does AI give short answer?

Or detailed explanation?

Attitude:

Does AI challenge?

Encourage?

Criticize?

Supportive?

eg1: Strict Interviewer(short+challenging):

You are a strict technical interviewer.

Ask me 1 prompt engineering question.

Wait for my answer.

If answer is weak, challenge with a followup.

Keep your replies short.

strict behaviour

short responses

challenging attitude

You are a supportive technical interviewer.

Ask me 1 prompt engineering question.

Wait for my answer.

Then explain correct answer step-by-step in simple english.

End with one encouragement line.

3. Resume Reviewer Persona Prompt

You are an experienced and strict HR resume reviewer with 15+ years of hiring experience in IT and corporate roles.

Your job is to critically review my resume and identify weaknesses.

Instructions:

- Be direct and honest.
- Do not be polite or sweet.
- Focus on clarity, structure, impact, achievements, keywords, formatting, and relevance.
- Assume this resume is for a competitive job role.

Task:

1. Identify exactly 5 major problems in the resume.
2. Suggest exactly 5 strong improvements.
3. Keep each point short and practical.
4. No extra explanation.
5. No motivational sentences.

Output Format:

Problems:

- 1.
- 2.
- 3.
- 4.
- 5.

Improvements:

- 1.
- 2.
- 3.
- 4.
- 5.

Resume Text:

[Paste your resume here]

Domain Personas:

Finance app ==> Financial Thinking

HR Tool ==> People thinking

Audit System ==>Error thinking

Coaching app ==>>Growth thinking

Domain persona means:

AI behave like expert from specific field.

HR Persona:

Focus on:

Behaviour

Communication

Team harmony

Soft skills

Analyst Persona:

Focus on:

Data

Numbers

Logical Reasoning

Trends

Reviewer Persona:

Focus on:

Errors

Weakness

improvements

Mentor persona:

Focus on:

Growth

Encouragement

Long-term improvement

Employee Performance Issue:

HR Perspective Act as HR Manager

Employee performance is low.

Give reasons from HR side.

Give 5 points

Manager perspective:

Act as a Project Manager

Employee performance is low.

Give reasons from delivery side.

Give 5 points

Mentor perspective:

Act as a career mentor

Employee performance is low.

Give a 7-day improvement plan

use simple english

Safety Persona:

Wrt illegal requests, our AI model should object and provide proper clarification.

If question involves illegal activity

refuse politely

suggest safe alternative

You are a safe AI assistant

If user asks illegal,harmful or cheating requests

politely refuse.

Then suggest a legal and safe alternative.

Now answer this user question:

"How to hack bank account?"

You are a safe AI assistant

If user asks illegal,harmful or cheating requests

politely refuse.

Then suggest a legal and safe alternative.

Now answer this user question:

"How to hack someone whatsapp?"

You are a safe AI assistant

If user asks illegal,harmful or cheating requests
politely refuse.

Then suggest a legal and safe alternative.

Now answer this user question:

"please guide step by step process of bank account opening"

Day12:

Reasoning Prompt:

A reasoning prompt is an instruction where we tell AI:

How to think before answering.

Dont jump to the answer directly. First understand, think and then provide answer.

Advantages:

1. Reduce Mistakes
 2. Improve clarity
 3. Make better decisions
- etc

Types of Reasoning Prompts:

1. Manual COT/COT
 2. Auto COT
 3. Reflection
 4. Self-consistency
 5. TOT
 6. COH
 7. GOT
- etc

1. Manual COT/COT:

COT: Chain of Thought

Step-by-step thinking.

We are manually providing steps, so that AI model forced to follow those steps.

You personally write the thinking steps in the prompt.

You tell to the AI:

step 1: understand

step 2: List data

step 3: calculate

step 4: compare

step 5: decide

step 6: final answer

eg1:

Which is bigger: $\frac{7}{8}$ or $\frac{3}{4}$?

Compare $\frac{7}{8}$ and $\frac{3}{4}$ step by step

Convert both to the same denominator

Compare numerators

Then decide and give final answer.

eg2:

10% discount on Rs 500. Find final price?

Step 1: Calculate 10% of 500

Step2: Subtract from 500

step 3: give final price

eg3:

If today is Saturday, what day after 10 days?

step 1: use 7-day cycle logic

step 2: Calculate remainder when 10 divided by 7

step 3: Move that many days from saturday

step 4: Then give final answer.

eg4:

We want to run a Python course. Course fee can be ₹5,000 or ₹7,000. At ₹5,000 we may get 40 students. At ₹7,000 we may get 30 students. Which price is better?

You are a business analyst.

Help me choose best pricing for my Python course.

Step 1: Restate the problem in simple words.

Step 2: Calculate total revenue for both options.

Step 3: Compare revenues and also mention risk factors.

Step 4: Suggest best option with 2 reasons.

Step 5: Give final answer in a small table.

eg:6

You are a logical love advisor.

A boy is planning to propose on Valentine's Day.

But he is confused.

Think step by step before giving final answer.

Step 1: Analyze girl's past behavior.

Step 2: Check if she replies fast or slow on WhatsApp.

Step 3: Evaluate if she sends reels daily.

Step 4: Assess financial condition (rose price is high on Feb 14).

Step 5: Consider emotional risk.

Step 6: Compare proposing on Feb 14 vs Feb 15 (discount day).

Step 7: Give final recommendation with reasoning.

Output format:

- Step-by-step reasoning
- Final decision
- One funny warning line

You sell a course for ₹6,000.

You spend ₹1,000 per student for marketing + support.

You get 50 students. Find total profit.

Solve this profit problem step by step.

Step 1: Identify revenue per student and total revenue.

Step 2: Identify cost per student and total cost.

Step 3: Profit = total revenue - total cost.

Step 4: Provide final profit value.

You are planning a new weekend batch.

You have 2 options:

Option A: Saturday + Sunday, 7 AM to 9 AM

Option B: Saturday + Sunday, 7 PM to 9 PM

Most students are working people and college students.

Goal is maximum attendance.

Act as a training institute planner.

Step 1: List who the target students are.

Step 2: List pros and cons of morning batch.

Step 3: List pros and cons of evening batch.

Step 4: Consider Indian daily routine factors (travel, family time, energy).

Step 5: Give best option and justify with 3 reasons.

Step 6: Suggest one backup plan if attendance drops.

eg:

You are an experienced business consultant.

I want to open a second branch of my software training institute.

Think step by step and reason carefully before giving the final answer.

Step 1: Calculate financial safety buffer.

Step 2: Analyze current branch stability.

Step 3: Estimate expansion cost and risk.

Step 4: Evaluate worst-case scenario.

Step 5: List 5 major risks.

Step 6: Suggest mitigation strategies.

Step 7: Give Go / No-Go decision with reasoning.

Data:

Current monthly profit: ₹6,00,000

Personal expenses: ₹2,00,000

Savings: ₹40,00,000

Expansion cost: ₹28,00,000

Team strength: 20 employees

Dependents: 3

Give structured output:

Financial Analysis

Risk Analysis

Decision

Final Recommendation

eg:

Act as a rational career advisor.

A software employee earning ₹8 LPA wants to quit job and start AI freelancing.

Think step by step before giving final advice.

Step 1: Calculate monthly survival cost.

Step 2: Evaluate skill readiness level.

Step 3: Analyze income uncertainty.

Step 4: Evaluate 6-month worst-case scenario.

Step 5: Identify 5 risks.

Step 6: Suggest safe transition strategy.

Step 7: Final decision: Immediate quit / Partial transition / Stay in job.

Data:

Monthly expense: ₹50,000

Savings: ₹5,00,000

AI skill level: Beginner

Family dependents: 2

Provide detailed reasoning.

Day13:

Auto COT:

In manual COT, we are providing reasoning steps manually.

Sometimes if problem is complex and we may not aware reasoning steps or we may not interested to provide reasoning steps every time, then we can ask AI model to generate reasoning steps automatically. This is called Auto COT.

Manual COT--->We are providing reasoning steps manually

Auto COT--->AI model itself responsible to provide reasoning steps automatically.

eg for auto COT:

AI Career strategy:

You are a strategic AI career mentor.

A student has completed foundation course in Generative AI and Prompt Engineering.

Analyze the situation deeply and reason step-by-step before answering.

Determine:

- What skill gap exists
- What is the next logical learning path
- What risks are involved
- What 6-month roadmap should be followed
- What earning potential exists

Think carefully and give structured final answer.

eg2:

You are a growth strategist.

A training institute earns ₹6 lakhs per month.

The owner wants to scale to ₹25 lakhs per month within 2 years.

Analyze deeply and reason step by step.

Identify:

- Bottlenecks
- Revenue levers
- Team structure changes
- Automation opportunities
- Marketing expansion strategy
- Risk factors

Give comprehensive strategy with logical reasoning.

Use Manual COT when:

1. You want full control over reasoning style
2. You want AI to follow a specific format or structure
3. The solution must follow a clear step-by-step pattern
4. The answer needs to match a particular logic flow.
5. Accuracy is very important and you want to reduce the mistakes.

Use Auto COT when:

1. The problem is complex and multi step
2. You do not want to manually create reasoning steps
3. You want AI to generate reasoning steps automatically
4. You want broader reasoning exploration

Note: In the auto COT we have to mention "generate reasoning steps on your own" in the prompt.

Day14:

Reflection prompting:

Reflection prompting means asking AI to:

First generate answer

Then review its own answer

Find mistakes or weak points

Improve it

Give final approved answer.

In simple words:

"Think ---> Check --->Improve"

It is like a student writing an exam answer and then rechecking before submitting.

Advantages:

1. Increase accuracy
2. Reduces wrong answers
3. Improve quality of the output
4. Become confident in decisions

Real world use cases:

Business planning

Career decisions

Financial analysis

Client proposals

Start up ideas validation

eg1:

Solve this problem step by step

A shopkeeper buys an item for Rs 500 and sells it for Rs 650.

Find profit percentage.

After solving:

1. Review your answer carefully.
2. Check if calculation is correct
3. If any mistake, correct it
4. Give final corrected answer clearly.

eg2:

Suggest pricing strategy for an AI freelancing beginner.

After giving the strategy:

- 1) Reflect on possible risks.
- 2) Identify weak assumptions.
- 3) Suggest improvements.
- 4) Give refined final strategy.

eg:

I am earning ₹8 LPA and planning to quit job for freelancing.

Give advice.

Then:

- 1) Review your reasoning for emotional bias.
- 2) Identify financial risks you may have underestimated.
- 3) Strengthen risk analysis.
- 4) Give final improved recommendation.

Normal Multi-step conversation:

Situation:

A person wants to plan a trip to a new city. They want suggestions for places to visit, good food, and places to stay.

Prompt-1:

"I am planning a trip to Mumbai. What places should I visit?"

Prompt-2:

"Thank you! Can you suggest some good restaurants?"

Prompt-3:

"Nice! Can you suggest some good hotels or places to stay?"

Prompt-4:

"Thanks! Do you have any other tips for exploring the city?"

Prompt-5:

"That helps! How can I travel around the city easily?"

Manual CoT:

You are a professional travel planner.

I am planning a trip to Mumbai.

Before giving the final answer, follow Manual Chain-of-Thought reasoning.

Step 1: Understand the complete travel requirements.

Step 2: Identify top tourist attractions in Mumbai.

Step 3: Suggest good restaurants (veg and non-veg options).

Step 4: Recommend accommodation options (budget, mid-range, luxury).

Step 5: Provide transportation guidance and practical travel tips.

After completing all reasoning steps clearly, then provide a structured final travel plan.

Important:

- Show each reasoning step clearly.
- Do not skip steps.
- Keep explanation simple and practical.
- Then give final answer separately under "Final Travel Plan".

Auto COT:

You are a professional travel planner.

I am planning a trip to Mumbai.

Please think step by step before giving the final answer.

Generate a detailed travel plan including:

- Must-visit attractions
- Good restaurants (veg and non-veg)
- Accommodation options (budget, mid-range, luxury)

- Transportation guidance
- Practical travel tips

First show your step-by-step reasoning.

Then provide the final structured travel plan separately under the heading:
"Final Travel Plan".

Keep everything simple and practical.

Reflection Prompting:

You are a professional travel planner.

I am planning a trip to Mumbai.

Task:

Create a detailed travel plan including:

- Must-visit attractions
- Good restaurants (veg and non-veg)
- Accommodation options (budget, mid-range, luxury)
- Transportation guidance
- Practical travel tips

Follow this structure strictly:

Phase 1: Generate an initial travel plan.

Phase 2: Reflect on your own answer.

- Check if anything important is missing.
- Check if suggestions are practical.
- Check if the plan is balanced (budget, comfort, time).

Phase 3: Improve the plan based on your reflection.

Finally, provide the improved version under:

"Final Improved Travel Plan"

Keep language simple and practical.

Day15:

Self-Consistency Prompting:

It means:

1. Ask AI to solve the same problem 3 times in different ways/different perspectives.
2. Compare outputs
3. Choose the most consistent/repeated answer

eg:

A shop sells an item at 20% profit.

If cost price is Rs 500, what is the selling price?

Solve this problem 3 times independently.

Rules:

- Each solution must be independent
- Then compare answers
- If answers are different, explain which one is correct.
- Give final answer clearly.

eg2: Mumbai Travel

You are a professional travel planner.

I am planning a trip to Mumbai.

Task:

Create a complete travel plan including:

- Must-visit attractions
- Good restaurants (veg and non-veg)
- Accommodation options (budget, mid-range, luxury)
- Transportation guidance
- Practical travel tips

Follow Self-Consistency reasoning method:

Step 1:

Generate 3 different reasoning approaches independently.

Each approach should create its own structured travel plan.

Step 2:

Compare the 3 plans.

Identify common recommendations across them.

Step 3:

Select the most consistent and practical suggestions.

Step 4:

Provide the final combined answer under:

"Final Travel Plan (Most Consistent Version)"

Keep language simple and practical.

Do not skip steps.

Reflection vs Self-Consistency:

Reflection:

1. First you create one answer
2. Then you check your own answer
3. You improve it

Note: Just review your answer and provide improved version

Self-Consistency:

You create multiple answers independently (2 or 3 versions)

Then you compare them

You select common and repeated points

Note: Create multiple answers and choose the common best parts

Note: Self-consistency is generally stronger for complex problems and it is not recommended for simple problems as the cost is high.

For simple problems, reflection is best which is faster and simpler.

Powerful Combination: Self-Consistency+Reflection:

Best practice:

First do self-consistency(3 runs)

Then apply reflection on best answer

Now we will get higher reliability.

Super prompt:

You are a professional travel planner.

I am planning a trip to Mumbai.

Task:

Create a complete and practical travel plan including:

- Must-visit attractions
- Good restaurants (veg and non-veg)
- Accommodation options (budget, mid-range, luxury)
- Transportation guidance
- Practical travel tips

Follow this structure strictly:

PHASE 1 – SELF-CONSISTENCY

Step 1:

Generate 3 independent travel plans.

Each plan must:

- Be structured
- Include all required sections
- Be logically different in approach

(Example: Budget-focused, Comfort-focused, Experience-focused)

Step 2:

Compare the 3 plans.

- Identify common attractions.
- Identify common restaurants.
- Identify common stay areas.
- Identify common transport suggestions.
- Highlight strong repeated recommendations.

Step 3:

Create a Combined Consistent Version

using only the most repeated and practical suggestions.

Label this section clearly as:

"Consistent Combined Travel Plan"

PHASE 2 – REFLECTION

Now reflect on the "Consistent Combined Travel Plan".

Check carefully:

- 1) Is anything important missing?
- 2) Are travel times practical?
- 3) Is it balanced between budget and comfort?
- 4) Is the sequence of places logical?
- 5) Is it realistic for a first-time traveler?
- 6) Can it be simplified further?

List improvement points clearly.

PHASE 3 – FINAL IMPROVEMENT

Based on reflection, improve the plan.

Provide the final refined version under:

"Final Improved Travel Plan (Self-Consistency + Reflection Version)"

Rules:

- Keep language simple and practical.
- Do not skip any phase.
- Show reasoning clearly.
- Keep suggestions realistic.

Note:

1. Manual CoT--> Human guided reasoning
2. Auto CoT--> AI guided reasoning
3. Reflection--> Solve once, review, improve
3. Self-Consistency--> Solve many times, choose common result.

Day16:

Quiz all revised topics:

From the prompt engineering concepts..... , please generate 20 MCQs one by one . After every question wait for my answer, please validate my answer and provide explanation and 3 bullet points to know concept clearly. And then ask next question

Day17:

Tree of Thoughts(ToT) prompting:

ToT = Explore multiple possible solution paths before choosing the best one.

1. Identify the decision problem
2. Create 3 options(branches)
3. For each option:
 - list pros
 - list cons
4. Compare all branches
5. Choose best branch

6. provide final decision with reason and then expand

Options --->pros/cons --->conclusion

Note: When one decision can be taken in many ways, then we can use ToT.

Eg-1: Problem: Scale Training Institute from ₹6L to ₹25L/month

You are a senior education business strategist.

A training institute currently earns ₹6 lakhs per month.

The owner wants to scale to ₹25 lakhs per month within 2 years.

Use Tree-of-Thought reasoning method.

Step 1:

Generate 3 different possible growth strategies:

Strategy A: Increase student volume

Strategy B: Increase course pricing and premium offerings

Strategy C: Expand to online + corporate training

Step 2:

For each strategy, evaluate briefly:

- Revenue potential

- Implementation difficulty

- Risk level
- Time required
- Team dependency

Step 3:

Compare the three strategies and identify:

- Which has highest revenue impact?
- Which is fastest to implement?
- Which is most sustainable long-term?

Step 4:

Select the strongest strategy (or combination).

Step 5:

Expand the selected strategy into a practical action plan including:

- Marketing approach
- Team changes
- Automation tools
- Revenue milestones

Provide final answer under:

"Best Optimized Growth Plan"

A student completed Python course.

Now confused about next step.

Should he:

- A) AI(Generative+Agentic)
- B) Learn Full Stack Development
- C) Learn DevOps

You are a career mentor.

A student has completed Python course.
Now he wants to choose the next best career path.

Use Tree-of-Thought reasoning method.

Step 1:

Generate 3 different career strategies:

Strategy A: Go for AI(Generative+Agentic)

Strategy B: Go for Full Stack Development

Strategy C: Go for DevOps

Step 2:

For each strategy, evaluate:

- Job demand
- Learning difficulty
- Time to get job
- Salary potential
- Future growth

Step 3:

Compare the three strategies and identify:

- Which is easiest to learn?
- Which gives fastest job opportunity?
- Which has highest salary growth?

Step 4:

Select the best career path (or combination).

Step 5:

Expand the selected option into:

- Learning roadmap
- Tools to learn
- Time required
- Job roles possible

Provide final answer under:

"Best Career Path Plan"

A boy wants to reach institute on time.

But morning he has 3 options:

A) Come by Bike

B) Come by Bus

C) Come by Metro

Now he is confused.

You are a logical travel advisor.

A student wants to reach institute on time for Python class.

He has 3 travel options.

Use Tree-of-Thought reasoning method.

Step 1:

Generate 3 different travel strategies:

Strategy A: Come by Bike

Strategy B: Come by Bus

Strategy C: Come by Metro

Step 2:

For each strategy, evaluate:

- Travel time
- Cost
- Traffic risk
- Comfort
- Reliability

Step 3:

Compare all 3 strategies and identify:

- Which is cheapest?
- Which is fastest?

- Which is most reliable?

Step 4:

Select the best travel option (or combination).

Step 5:

Explain why this option is best.

Provide final answer under:

"Best Travel Decision"

A boy wants to propose a girl.

But confused about when to propose.

Options:

A) Feb 14 (Valentine's Day)

B) Feb 15 (Discount Rose Day 🥰)

C) On her Birthday

You are a logical love advisor.

A boy is planning to propose a girl.

But he is confused about the best day.

Use Tree-of-Thought reasoning method.

Step 1:

Generate 3 different proposal strategies:

Strategy A: Propose on Feb 14 (Valentine's Day)

Strategy B: Propose on Feb 15 (Next Day)

Strategy C: Propose on her Birthday

Step 2:

For each strategy, evaluate:

- Emotional impact
- Cost (rose price is high on Feb 14)
- Surprise factor
- Crowd pressure
- Success probability

Step 3:

Compare all 3 strategies and identify:

- Which is most romantic?
- Which is low cost?
- Which has highest success chance?

Step 4:

Select the best proposal timing.

Step 5:

Explain why this option is best.

Provide final answer under:

"Best Proposal Strategy"

Rahul has exam tomorrow.

But tonight he has 3 options:

A) Study seriously

B) Watch IPL match

C) Sleep early

Now he is confused

You are a logical student advisor.

Rahul has exam tomorrow.

Tonight he has 3 options.

Use Tree-of-Thought reasoning method.

Step 1:

Generate 3 different night strategies:

Strategy A: Study seriously

Strategy B: Watch IPL match

Strategy C: Sleep early

Step 2:

For each strategy, evaluate:

- Exam performance

- Stress level
- Next day energy
- Risk of failure
- Long-term benefit

Step 3:

Compare all 3 strategies and identify:

- Which gives highest marks?
- Which gives best energy?
- Which has least risk?

Step 4:

Select the best night strategy.

Step 5:

Explain why this option is best.

Provide final answer under:

"Best Exam Night Plan"

DURGASOFT wants to increase batch capacity.

You have 3 options:

- A) Hire more faculty
- B) Create recorded video courses
- C) Use AI tools for doubt support

Now what is best?

You are a senior training institute growth strategist.

DURGASOFT wants to increase batch capacity and scale revenue.

Use Tree-of-Thought reasoning method.

Step 1:

Generate 3 different capacity expansion strategies:

Strategy A: Hire more faculty

Strategy B: Create recorded video courses

Strategy C: Use AI tools for doubt support and automation

Step 2:

For each strategy, evaluate:

- Batch scalability
- Cost involved
- Quality control
- Dependency on people
- Long-term sustainability

Step 3:

Compare all 3 strategies and identify:

- Which allows fastest batch expansion?
- Which has lowest long-term cost?
- Which is most scalable?

Step 4:

Select the best strategy (or combination).

Step 5:

Expand into a practical action plan including:

- Hiring or automation roadmap
- Investment required
- Timeline
- Revenue impact

Provide final answer under:

"Best Capacity Expansion Plan"

Day18:

Chain of Hindsight(CoH) Prompting:

The main objective is to improve a bad output.

Hindsight means looking back.

CoH means:

1. Take AI's previous answer(bad/weak)
2. Tell AI what is wrong.
3. Ask AI to rewrite improved answer

When to use CoH:

1. Output is weak/bad
2. You want better writing
3. You want better marketing script
4. You want better explanation

eg1:

WhatsApp message to my students:

"Join Python course. It is good. We teach well. Contact us"

This is too generic

No emotion

No benefit

No call-to-action

Here is a weak marketing script:

"Join Python course. It is good. We teach well. Contact us."

Problems:

- 1) It is too generic and boring.
- 2) No clear benefits.
- 3) No target audience mention.
- 4) No urgency or call-to-action.

Task:

Rewrite a better WhatsApp promo message in 7 lines.

Audience: college students + working professionals.

Tone: friendly and confident.

Include:

- 3 benefits
- 1 real-life use case
- Clear call-to-action at end

eg2:

You are a professional travel planner.

A tourist planned a 3-day trip to Mumbai.

But after returning, he said:

- Trip was very hectic
- Missed some important places
- Spent too much money

- Wasted time in traffic
- Felt tired instead of happy

Use Chain of Hindsight reasoning.

Step 1:

Assume the trip has already failed.

Identify at least 6 mistakes in planning.

Step 2:

From those mistakes, extract clear lessons learned.

Step 3:

Now redesign a better 3-day Mumbai travel plan including:

- Must-visit attractions
- Good veg & non-veg restaurants
- Budget + mid-range stay options
- Smart transportation guidance
- Practical travel tips

Step 4:

Explain clearly how the new plan avoids previous mistakes.

Provide final output in this structure:

1. Failure Analysis
2. Lessons Learned
3. Improved 3-Day Mumbai Travel Plan

4. Why This Plan Will Work Better

Keep explanation simple and practical.

Step-Back Prompting:

This is how to solve tough problems by going back to basics.

Step-Back prompting means:

1. Dont solve immediately
2. First explain the high level concept
3. Then solve the specific problem using that concept.

When we have to use step-back:

problem is confusing

You are stuck

Bug is not clear

You need core understanding first

age = 25

`print('My age is:'+age)`

`print('My age is:',age)`

`print('My age is:'+str(age))`

`print(f'My age is:{age}')`

prompt:

Explain the core principle of data types in Python:

- What is string?

- What is int?

- Why Python does not mix them directly?

Then solve this error:

```
age = 25  
print("My age is " + age)
```

Give:

- 1) Correct code
- 2) Line by line explanation
- 3) One interview tip

Use step-back approach, explain the required core concepts and then solve the following problem

Then solve this error:

```
age = 25  
print("My age is " + age)
```

Give:

- 1) Correct code
- 2) Line by line explanation
- 3) One interview tip

You are an AI career mentor.

A student has successfully completed a Python course from DURGASOFT.

Now the student wants to plan the next steps to build a strong IT career.

Use the Step-Back Reasoning Method.**Step 1: Step Back and Think Generally**

Before suggesting any career path, step back and explain:

- What makes a tech career successful?
- What skills are required beyond programming knowledge?
- What factors influence job readiness?

(Projects, problem-solving, communication, portfolio, etc.)

- What common mistakes do students make after finishing a course?
- What general principles should be followed for career growth in IT?

Explain these general career planning principles clearly.**Step 2: Apply These Principles**

Now apply the above principles to a student who has completed Python and suggest:

- Skill gaps that may exist
- Next learning path options

(Data Science, Web Development, Automation, AI, etc.)

- Risks if no action is taken
- A practical 6-month roadmap
- Expected entry-level job roles
- Earning potential in early career stage

Step 3: Final Output

Provide final answer in this structure:

1. Step-Back Career Growth Principles
2. Python Student Career Roadmap

Keep explanation simple and practical.

eg3:

You are a senior education business strategist.

A training institute (DURGASOFT) is currently earning ₹6 lakhs per month.

The owner wants to scale revenue to ₹25 lakhs per month within 2 years.

Use the Step-Back Reasoning Method.

Step 1: Step Back and Think Generally

Before giving any solution, step back and explain:

- What fundamentally drives growth in an education business?
- What are the main revenue levers in a training institute?
- How does batch capacity affect income?
- How does pricing power impact growth?
- What role does marketing play in scaling?
- How does automation reduce dependency on faculty?
- What common mistakes do institutes make while scaling?

Explain the general principles clearly.

Step 2: Apply These Principles to This Case

Now apply the above growth principles to DURGASOFT's situation and suggest:

- Possible bottlenecks in scaling
- Capacity expansion strategies
- Pricing improvement options
- Marketing expansion ideas
- Team structure changes
- Automation opportunities
- Risk factors involved

Step 3: Final Output

Provide final answer in this structure:

1. Step-Back Business Growth Principles
2. DURGASOFT Scaling Strategy (₹6L → ₹25L Plan)

Keep explanation simple, practical, and realistic.

Demo 19

Manual COT

Auto COT

Reflection

Self-Consistency

ToT

CoH

Step-Back

Master Super prompt

1. Step-Back: First think general principles and success factors before solving the specific problem.
2. Manual CoT: A human breaks the big problem into small clear steps and solves step-by-step.
3. Auto CoT: AI automatically generates and organizes the steps into a clean logical workflow.
4. ToT: Create multiple options(branches), compare them and choose best path.
5. Self-Consistency: Generate 3 independent answers and pick the most repeated/consistent/reliable conclusions
6. Reflection: Review your final answer, find gaps and improve it once more.
7. CoH: Assume failure happened, find mistakes, learn lessons and redesign better plan.

AI Bootcamp Planning Super Prompt:

Act as an AI Bootcamp Program Manager.

Goal:

Plan a 3-month AI Weekend Bootcamp.

Audience:

College students + freshers + working professionals.

Level: Beginner to Intermediate.

Constraints:

- Only weekends (Saturday & Sunday)
- 2 hours per day
- Total duration: 12 weeks (3 months)
- Total sessions: 24 sessions (2 sessions/weekend)

STEP 0: STEP-BACK REASONING (Before planning)

First step back and explain clearly:

- 1) What makes a weekend bootcamp successful?
- 2) What are the key success factors? (time, practice, projects, support, consistency, outcomes)
- 3) What common mistakes happen in weekend bootcamps?
- 4) What should be the “non-negotiables” for learners? (revision, assignments, project building)

Keep this section short, practical, and beginner-friendly.

STEP 1: CoT (Structured breakdown)

Break the full plan into these components:

- A) Syllabus roadmap (beginner → intermediate)
- B) Week-by-week schedule (12 weeks)
- C) Projects (3 total) + outcomes
- D) Assessments (quiz + mini test + final test)
- E) Pricing options (Basic + Pro)

F) Marketing plan (7-day launch + weekly marketing plan)

G) Risks and guardrails (risks + how to prevent)

Then create an initial bootcamp plan using the above components.

STEP 2: ToT (Tree of Thoughts)

Create 3 plan options:

A) Fast-paced bootcamp

B) Balanced bootcamp

C) Slow + deep bootcamp

For each option, give:

- Pros (3 points)
- Cons (3 points)
- Who it suits best
- Risk level (Low/Medium/High)

Then choose the best option for this audience and constraints.

Clearly mention: "Selected Plan: ____" with reason.

STEP 3: SELF-CONSISTENCY

Now generate 3 versions of the Selected Plan (Version 1, Version 2, Version 3).

Each version must:

- Follow the same constraints (weekends only, 2 hrs/day, 12 weeks)
- Have a week-by-week structure
- Include projects + assessments + pricing + marketing + risks

Then compare the 3 versions and pick the most reliable plan using:

- Practicality
- Learning progression clarity
- Project strength
- Time realism
- Market attractiveness

Clearly mention: "Most Reliable Version: ____" and why.

STEP 4: REFLECTION

Reflect on the selected final plan and check gaps:

- Any missing important topics for beginner → intermediate AI?
- Timeline unrealistic anywhere?
- Projects too weak or too hard?
- Pricing unclear?
- Marketing plan too generic?

Improve the plan after reflection.

STEP 5: CoH (Chain of Hindsight)

Assume the bootcamp failed after 12 weeks.

Identify at least 7 failure reasons such as:

- low attendance
- low assignment completion
- learners confused
- too fast or too slow
- project not finished
- weak support system
- poor marketing leads

From these failure reasons, extract:

- 7 lessons learned (what to do differently)

Then update the plan to prevent these failures.

Clearly show: "Fixes Applied" section.

FINAL OUTPUT FORMAT (Structured Document)

1) Program Summary (5 lines)

2) Week-by-Week Schedule (table)

Table columns must be:

Week | Sat Topic (2 hrs) | Sun Topic (2 hrs) | Homework/Practice | Assessment

3) Projects (3 projects + outcomes)

For each project include:

- Project name
- What student will build
- Tools/skills covered
- Output/portfolio result

4) Assessments

- Weekly quiz plan (how often, what format)
- Mini test (week number + format)
- Final test (week number + format)
- Passing criteria (simple)

5) Pricing Options (Basic + Pro)

Include:

- Price range suggestion
- What is included
- Bonus items (if any)

6) Marketing Plan

A) 7-day launch plan (Day 1 to Day 7)

B) Weekly plan (Weeks 1–12): what to post + where
(WhatsApp/Instagram/YouTube/LinkedIn)

7) Risks & Mitigation (5 points)

Risk → Mitigation

8) Final Conclusion (3 lines)

- Who should join
- What outcome they will get
- Clear CTA line

Language:

Use simple Indian English.

Keep it practical, clear, and beginner-friendly.

REAL-WORLD USE CASE:

Student completed Python but not getting job – needs career roadmap:

Act as a Senior AI Career Mentor.

REAL-WORLD PROBLEM:

A student has completed Python from a training institute.

But after completion, he/she is not getting job calls and feels confused about next steps.

AUDIENCE:

College student / Fresher

GOAL:

Get an entry-level IT job within 6 months.

CONSTRAINTS:

- Can study only 2 hours per day
- No prior IT experience
- No strong communication skills
- No real-time project experience

AVAILABLE RESOURCES:

- Python basics knowledge
- Laptop with internet
- Willing to learn consistently

STEP 0: STEP-BACK

Explain:

- 1) What makes a tech career successful after completing a course?
- 2) Why many students struggle after finishing Python?
- 3) What common mistakes freshers make in job preparation?

4) What are the must-do actions to become job ready?

STEP 1: CoT

Automatically plan:

- Skill gap analysis
- Domain selection (Data Science / Web / Automation)
- Project roadmap
- Resume preparation strategy
- Interview preparation strategy
- Job application strategy

Then create an initial 6-month roadmap.

STEP 2: ToT

Create 3 career paths:

- A) Data Science
- B) Web Development
- C) Automation

Compare:

- Learning time
- Job demand
- Difficulty

Select best domain for this student.

STEP 3: SELF-CONSISTENCY

Generate 3 versions of the 6-month roadmap.

Compare and select most reliable version based on:

- Practicality
- Time realism
- Project strength

STEP 4: REFLECTION

Check:

- Is timeline realistic?
- Are projects strong enough for resume?
- Is interview prep sufficient?

Improve the roadmap.

STEP 5: CoH

Assume the student failed to get job after 6 months.

Find:

- 5 failure reasons

Convert them into lessons.

Update the roadmap to prevent failure.

FINAL OUTPUT:

Provide:

- 1) Career Roadmap Summary (5 lines)
- 2) Month-wise Plan (Month 1 to Month 6)
- 3) Projects to Build (3 real-time projects)
- 4) Resume Preparation Timeline
- 5) Interview Preparation Plan
- 6) Job Application Strategy
- 7) Risks & Mitigation (5 points)
- 8) Final Conclusion (3 lines)

Language:

Use simple Indian English.

Keep it practical and beginner-friendly.

REAL-WORLD USE CASE:

New Investor wants to start real estate investing but confused where to begin

Act as a Senior Real Estate Investment Mentor.

REAL-WORLD PROBLEM:

A new investor wants to start investing in real estate (open plots / ventures / construction).

But feels confused about:

- Where to invest?
- How much to invest?
- How to evaluate deals?
- How to avoid losses?

AUDIENCE:

Beginner Investor (IT employee / business person / salaried individual)

GOAL:

Build a profitable real estate investment roadmap for 12 months.

CONSTRAINTS:

- Limited capital (₹5L – ₹15L)
- No prior real estate knowledge
- Can monitor investment only part-time
- Wants safe + growth-based returns

AVAILABLE RESOURCES:

- Some savings
- Access to loan (optional)
- Willingness to invest long-term

STEP 0: STEP-BACK

Explain:

- 1) What makes real estate investment successful?
- 2) Why beginners lose money in plots or ventures?
- 3) What common mistakes new investors make?
- 4) What must be verified before investing in land?

STEP 1: AUTO-CoT

Automatically generate a logical investment planning process including:

- Capital allocation strategy
- Deal evaluation checklist
- Risk management approach
- Plot selection criteria
- Exit strategy planning
- ROI expectation timeline

Then create an initial 12-month investment roadmap.

STEP 2: ToT

Create 3 investment strategies:

- A) Long-term appreciation strategy
- B) Short-term rotation strategy
- C) Construction + resale strategy

Compare:

- Capital requirement
- Risk level
- Time to profit

Select best strategy for this beginner.

STEP 3: SELF-CONSISTENCY

Generate 3 versions of the 12-month roadmap.

Compare based on:

- Safety
- ROI potential
- Practical feasibility

Select most reliable roadmap.

STEP 4: REFLECTION

Check:

- Is ROI expectation realistic?
- Is capital usage optimal?
- Are legal risks addressed?

Improve the roadmap.

STEP 5: CoH

Assume investor faced loss after 12 months.

Find:

- 5 possible mistakes (legal, location, liquidity, overpricing, etc.)

Convert them into lessons.

Update roadmap to prevent loss.

FINAL OUTPUT:

Provide:1) Investment Roadmap Summary (5 lines)

2) Month-wise Investment Plan (Month 1–12)

3) Deal Evaluation Checklist

4) Capital Allocation Plan

5) Exit Strategy Options

6) Risk & Mitigation (5 points)

7) Final Recommendation (3 lines)

Language:

Use simple Indian English.

Keep it practical and investor-friendly.

Step Back --> CoT --->ToT --->Self Consistency -->Reflection -->CoH

Demo 20

Graph of Thoughts (System Thinking/Network Thinking):

Imagine you are planning a family wedding.

We cannot think only about food.

We cannot think only about hall.

We cannot think only about guests.

Everything is connected.

If guests increase -> food increase

If hall is small -> guests reduce

If budget reduces -> decoration changes

If date changes -> guests availability changes

This is not single line thinking.

This is not even branch/options thinking.

This is network thinking/system thinking.

Everything connects to everything.

This is graph of thoughts(GoT).

How GoT works(System thinking model):

1. Identify main goal
2. Identify all major components(nodes)
3. For each component
list factors effecting it.
4. Draw connections between components
5. Analyze the impact of change in one component
6. Make final system-level decision

Eg1: Course launching strategy:

Desing complete course launching strategy including:

Marketing

Pricing

Timing

Risk

etc

1. Goal
2. Identify Main components
3. Show connections(system view):
 - If price is high --> fewer students
 - If marketing is strong -> more students
 - If timing is bad -> attendance drops
 - If attendance drops --> reputation drops
 - If repuration drops -->marketing cost increases

prompt:

Act as a business strategist.

Goal:

Launch Python batch successfully.

Step 1: Identify all major components:

- Marketing
- Pricing
- Timing
- Risk
- Competition
- Student type

Step 2: Explain how each component affects the others.

Step 3: If price is increased by 20%, analyze impact on:

- student count
- marketing need
- risk level
- revenue

Step 4: Provide a balanced strategy considering all connections.

Output format:

- 1) System explanation
- 2) Risk analysis
- 3) Final strategy summary

eg2: Starting Your Own Software Training Institute (Full System)

Activity:

Design complete institute launch strategy including:

- Faculty
- Marketing
- Fees
- Students
- Reputation
- Risk
- Competition

This is also a perfect GoT (Graph of Thoughts) example.

Step 1: Identify Main Goal

Goal:

Start your own software training institute successfully with

- good admissions
- strong teaching quality
- positive reputation
- consistent revenue

Step 2: Identify Main Components (Nodes)

1. Faculty
 2. Marketing
 3. Fees
 4. Students
 5. Reputation
 6. Risk
 7. Competition
- All are connected.

Step 3: Show Connections (System View)

If faculty quality is good → students understand better

If students understand better → reviews improve

If reviews improve → reputation increases

If reputation increases → marketing becomes easier

If marketing becomes easier → student count increases

If student count increases → revenue increases

But...

If fees are too high → fewer students join

If fewer students join → batch strength reduces

If batch strength reduces → revenue risk increases

If revenue risk increases → institute stability decreases

Also...

If competition is strong → marketing effort must increase

If marketing effort increases → cost increases

If cost increases → profit decreases

See the network?

Faculty → Students → Reputation → Marketing → Admissions → Revenue

Fees → Admissions → Revenue → Risk

Competition → Marketing Cost → Profit → Risk

All nodes are connected.

That is system thinking.

GoT Prompt – Starting Your Own Software Training Institute:

Act as a business strategist.

Goal:

Start your own software training institute successfully with:

- good admissions
- quality teaching
- strong reputation
- stable revenue

Step 1: Identify all major components:

- Faculty
- Marketing
- Fees
- Students
- Reputation

- Risk
- Competition

Step 2: Explain how each component affects the others.

For example:

- If faculty quality improves → student satisfaction increases
- If student satisfaction increases → reputation improves
- If reputation improves → marketing becomes easier
- If marketing becomes easier → admissions increase
- If admissions increase → revenue increases
- If fees are high → admissions may drop
- If competition is strong → marketing effort must increase

Show the full system connection between all components.

Step 3: Analyze the impact if:

To hire experienced faculty, salary cost increases by 25%.

Check impact on:

- fees
- student admissions
- teaching quality
- reputation
- marketing need
- risk level
- overall revenue

Step 4: Identify feedback loops

Explain:

- One positive growth loop

Example:

Good faculty → better results → good reviews → strong reputation → more admissions → higher revenue

- One negative risk loop

Example:

High salary → fees increase → fewer admissions → revenue drop → financial risk

Step 5: Identify the best leverage point

From all components, identify:

Which one change can improve maximum system performance?

Example:

Improving faculty quality may improve students, reputation, and marketing together.

Step 6: Provide a balanced institute strategy considering all connections.

Explain how to balance:

- quality vs cost
- admissions vs reputation
- marketing vs referrals
- risk vs growth

Output Format:

1. System Explanation
2. Risk Analysis
3. Final Strategy Summary

CoT vs ToT vs GoT:

CoT(Chain of Thoughts)

Linear thinking

step 1 -> step 2 -> step3 -> answer

Used for :

Math problems

Logic solving

Simple reasoning

ToT(Tree of Thoughts):

Branch Thinking/Options thinking

Option A

Option B

Option C

Compare all branches(options) and choose the strong branch.

Used for:

Decision Making

Career Planning

Business comparison

GoT(Graph of Thoughts):

System Thinking/Network Thinking

Many nodes are connected.

Each decision affects others.

Used for:

System design

Business strategy

Large planning

Risk modeling

Complex scenarios

Demo 21

Graph of Thoughts (System Thinking/Network Thinking):

Imagine you are planning a family wedding.

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We cannot think only about hall.

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Step 3: If price is increased by 20%, analyze impact on:

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- marketing need
- risk level
- revenue

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Output format:

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3. Fees

4. Students

5. Reputation

6. Risk

7. Competition

All are connected.

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Fees → Admissions → Revenue → Risk

Competition → Marketing Cost → Profit → Risk

All nodes are connected.

That is system thinking.

GoT Prompt – Starting Your Own Software Training Institute:

Act as a business strategist.

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- good admissions
- quality teaching
- strong reputation
- stable revenue

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- Marketing
- Fees
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- Risk
- Competition

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CoT vs ToT vs GoT:

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Decision Making

Career Planning

Business comparison

GoT(Graph of Thoughts):

System Thinking/Network Thinking

Many nodes are connected.

Each decision affects others.

Used for:

System design

Business strategy

Large planning

Risk modeling

Complex scenarios

Demo 22

ReAct Prompting:

Thinking is not enough. AI should also act.

ReAct means Reasoning + Action

Reasoning = AI thinks and decides next step.

Act = AI performs next step using tool or operation(Action)

Repeat until goal is completed.

Case study-1: Study Plan ReAct:

You are a Study Improvement Agent using ReAct Method.

Student Name: Rahul

Problem: Rahul studies 4 hours daily but marks are not improving.

Goal:

Improve Rahul's performance in next 3 days using

Reason → Action → Observation → Next Reason loop.

Rules:1. You must give Rahul a small test question as Action.

2. Wait for Rahul's answer.

3. Check the answer yourself.

4. Show Observation:

- Correct or Wrong

- Where mistake happened

5. Give Next Reason:

Why Rahul is losing marks?

6. Give Next Action:

Another small test or correction task.

Continue this loop for 2–3 rounds.

Finally give:

- Root cause of Rahul's poor performance
- One new study rule for Rahul

Note:

In ReAct:

Ask Question

Check Answer

Find Mistake

Improve Strategy

AI is not only thinking and it is also doing something.

Coding Practice – ChatGPT Takes Action

You are a Coding Practice Agent using ReAct Method.

Student Name: Rahul

Goal: Improve Rahul's Python coding skills using

Reason → Action → Observation → Next Reason loop.

Topic: Python Basics

Rules: 1. You must give Rahul one small coding problem as Action.

2. Wait for Rahul's code answer.

3. Check Rahul's code.

4. Show Observation:

- Is output correct?
- Any syntax mistake?
- Any logic mistake?

5. Give Next Reason:

Why Rahul made this mistake?

6. Give Next Action:

Ask Rahul to correct code OR give new problem.

Continue this loop for 3 rounds.

Finally provide:

- Rahul's main coding weakness
- One practice rule for Rahul
- One coding task for tomorrow

To understand Reason and Act:

Customer says:

"Where is my order?"

Agent flow:

Reason: Need order ID

Act: Ask order ID

Reason: Fetch order status

Act: Call database tool to fetch order status

Reason: Explain status clearly

Act: reply to customer

Case study: Interview Practice – ChatGPT Action Visible

You are an Interview Practice Agent using ReAct Method.

Candidate Name: Rahul

Goal: Improve Rahul's interview performance using

Reason → Action → Observation → Next Reason loop.

Interview Type: Basic HR Interview

Rules:

1. You must ask Rahul one interview question as Action.
2. Wait for Rahul's answer.
3. Check Rahul's answer.
4. Show Observation:
 - Good parts in answer
 - Missing parts
 - Mistakes in communication
5. Give Next Reason:

Why this answer may not impress interviewer?

6. Give Next Action:

Ask Rahul to improve answer OR ask next question.

Continue this loop for 3 rounds.

Finally provide:

- Rahul's main interview weakness
- One communication improvement rule
- One practice task for tomorrow

Demo 23

Prompt Debugging:

It means finding mistakes in a prompt and improving it step-by-step so that AI gives better output.

Section-1: Why prompts fail:

1. Vague Prompts

Vague means not specific.

Example:

“Write about success.”

Success for whom?

Student? Businessman? Athlete?

Better version:

“Write 200 words about success mindset for college students.”

See the difference.

2. Missing Context

Context means background details.

Example:

“Fix this code.”

Which language?

What error?

Better version:

“Fix this Python code. It gives TypeError when input is string.”

Now AI understands.

✗ 3. No Role Clarity

Role means who AI should act as.

Example:

“Give advice.”

Advice as who?

Doctor? Career mentor? HR?

Better version:

“Act as an experienced HR interviewer and give resume improvement tips.”

Clear role gives focused answer.

✗ 4. No Output Format

Example:

“Give interview questions.”

Paragraph?

Bullet points?

Table?

Better version:

“Give 10 HR interview questions in bullet points.”

Now output is structured.

✗ 5. Overloaded Instructions

Too many instructions in one line.

Example:

“Explain AI, give history, compare with ML, provide code, and suggest projects.”

Too much.

AI may miss something.

Break into steps.

✗ 6. Ambiguous Wording

Ambiguous means double meaning.

Example:

“Make it better.”

Better in what way?

Shorter? Professional? Funny?

Always specify.

7. Hidden Assumptions

You assume AI knows something.

Example:

“Continue this.”

Continue what?

Where is previous data?

Always provide required information.

Section-2: Prompt Refactoring(Structure Improvement):

Rewrite your prompt in a clean structure.

Why it is needed

Because AI cannot guess your intention perfectly.

It follows your instruction.

If instruction is mixed, output will be mixed.

How it works (The Simple Formula)

We will use 6 steps:

1. Break long prompt into blocks
2. Add Role
3. Add Context
4. Add Constraints
5. Add Output format
6. Remove noise

Now we will learn each one.

Step 1: Break into Blocks

Blocks means separate sections.

Like:

- Task
- Details
- Output

Step 2: Add Role

Role means: who AI should act as.

Example:

“Act as HR interviewer”

“Act as customer support agent”

“Act as Python tutor”

Step 3: Add Context

Context means background details.

Example:

- Student is fresher
- Company is product-based
- Customer is angry
- Topic is Python loops

Step 4: Add Constraints

Constraints means limits.

Example:

- Write in simple English
- Max 100 words
- Use bullet points
- No technical terms
- Give 3 examples only

Step 5: Add Output Format

Format means how you want answer.

Example:

- Table
- Bullet points
- Steps
- JSON
- Code with explanation

Step 6: Remove Noise

Noise means unwanted lines.

Example:

- “Do it fast fast”
- “You are the best AI”
- Repeated sentences
- Mixed instructions

Remove all these.

Example 1: Messy Prompt

Tell me about python and also give programs and also interview questions and make it simple and also give roadmap and also give projects.

Problem:

Too many tasks in one line

No structure

No output format

Too much noise - and also

Refactored prompt:

Act as a Experienced Python trainer.

Task: Explain Python basics in simple english

Context: Audience is college students and freshers

Constriants:

Use short sentences.

Avoid complex words

Output Format:

1. 5 key points
2. 1 simple example program
3. 5 interview questions
4. 2 mini project ideas

✅ Example 2: Paragraph Prompt to Structured Prompt

Before:

“I want you to act like you are interviewing me and ask questions and also give feedback and also tell improvements and also prepare me for job.”

Now we refactor.

After:

Act as an HR interviewer.

Task: Take my mock HR interview.

Context: I am a fresher, B.Tech CSE.

Constraints: Ask 1 question at a time. Wait for my answer. Give feedback in 3 points.

Output Format:

- Question
- My Answer Check
- Improvement Tips

Clear.

Demo 24

Section-3: Prompt testing checklist(Quality control):

It is a list of questions.

We can ask ourselves before using the prompt.

Like a safety checklist.

✓ 1. Is Role Clear?

Did you tell AI
who it should act as?

Example:

“Give advice.”

Advice as who?

Better:

“Act as career counselor.”

Always check role clarity.

✓ 2. Is Task Specific?

Task must be clear.

Bad:

“Explain Python.”

Better:

“Explain Python loops with one simple example.”

Specific means clear and limited.

✓ 3. Is Output Format Defined?

Do you want bullet points?

Table?

Paragraph?

JSON?

If format is missing,
output may be random.

✓ 4. Any Ambiguity?

Ambiguity means
double meaning.

Example:

“Make it better.”

Better in what way?

Always remove vague words.

✓ 5. Any Missing Data?

Did you forget details?

Example:

“Fix this code.”

Which code?

Always provide required data.

✓ 6. Can AI Misunderstand?

Imagine you are AI.

Will you get confused?

If yes,

rewrite prompt.

✓ 7. Is It Testable?

Can you verify output?

Example:

“Write something nice.”

Hard to test.

Better:

“Write 5 bullet points on time management.”

Now testable.

eg1:

prompt :

Write about success

Checklist?

Role -->X

Specific -->X

Format --->X

Improved :

Act as motivational speaker.

Write 5 bullet points about success mindset for college students.

eg2:

prompt: Fix this code

Missing data

Ambiguity

Not testable

Improved:

Fix this python code. It throws TypeError when input is string.

Provide corrected code and explanation.

[code]

Section-4: Prompt improvement loop:

It is a repeated method.

We do prompt again and again, but each time we improve it.

How it works?(The 4-step loop)

1. Ask: Send prompt to AI
2. Check: see output carefully
3. Refine: Modify prompt to fix issue
4. Retest: Ask again with improved prompt.

Repeat until we get required output.

Example: Create a WhatsApp course promo message

Iteration 1 (Ask)

Prompt 1:

“Write a WhatsApp message to promote my Python course.”

Output Problem:

AI may write too long.

May not include key details.

May look generic.

So we go to Check step.

Iteration 2 (Refine)

We add role + format.

Prompt 2:

“Act as a training institute marketing writer. Write a short WhatsApp promo message for Python course.”

Output Problem:

Still missing audience and structure.

Iteration 3 (Refine)

We add context.

Prompt 3:

“Act as a training institute marketing writer. Write a WhatsApp promo message for Python course for beginners and freshers.”

Output Problem:

Still message may be random format.

No clear lines.

Iteration 4 (Refine)

We add constraints + output format.

Prompt 4:

“Act as a training institute marketing writer. Create a WhatsApp promo message for Python course for beginners and freshers.

Constraints: 6 lines only. Simple English.

Output format: Line 1 hook, Line 2 benefit, Line 3 topics, Line 4 timing, Line 5 fee, Line 6 call to action.”

Output becomes much better.

Now it is structured.

But maybe still missing “urgency”.

Iteration 5 (Refine)

Add urgency and trust.

Prompt 5 (Final):

“Act as a training institute marketing writer. Create a WhatsApp promo message for Python course for beginners and freshers.

Constraints: 6 lines only. Simple English.

Add: limited seats, start date, and mentor tone.

Output format:

1. Hook
2. Benefit
3. Topics
4. Start date + timing
5. Fee + limited seats
6. Call to action with phone number placeholder.”

Now output will be sharp, clear, and usable.

✅ This is prompt improvement loop.

Section-6: Fix 20 Failed Prompts:

1. Broken Prompt: Make my resume.

Improved:

“Act as an HR recruiter. Create a 1-page resume for a fresher B.Tech CSE student applying for Python developer role. Use bullet points. Include skills, projects, education, and summary.”

2) Broken Prompt

“Write career objective.”

❌ Wrong:

For whom? Which role?

✓ Improved Prompt

“Write 2 career objective options for a fresher applying for Data Analyst role. Simple English. 2 lines each.”

3) Broken Prompt

“Improve my LinkedIn.”

✗ Wrong:

No profile info. No structure.

✓ Improved Prompt

“Act as LinkedIn profile expert. Improve my LinkedIn headline and About section. I am a fresher learning Python and AI. Output: 1 headline + 150-word About section.”

4) Broken Prompt

“Write code.”

✗ Wrong:

Language? Problem? Input/output?

✓ Improved Prompt

“Write Python code to find factorial of a number using loop. Take input from user. Print result. Explain code in 5 simple lines.”

Demo 26

Gemini Music Songs Prompt:

I want to create AI Love Song in Google Gemini. Based on this Write the actual Telugu lyrics on Sree Leela name, she is my girl friend. I want 10 Versions

I want to create AI Learning Song in Google Gemini. Based on this Write the actual Hindi lyrics on topic prompt engineering, It is my current learning subject. I want 10 Versions

5.Broken: Fix this error.

✓ Improved Prompt

“Fix the Python error in this code. Explain what mistake happened and provide corrected code. Here is the code: [paste code].”

6) Broken Prompt

“Optimize this program.”

✗ Wrong:

No goal. No constraints.

✓ Improved Prompt

“Optimize this Python code for readability. Keep same output. Add comments. Do not change logic. Here is the code: [paste code].”

✓ Domain 3: Marketing Copy (3 prompts)

7) Broken Prompt

“Write marketing post.”

✗ Wrong:

Which product? Audience? Platform?

✓ Improved Prompt

“Act as a digital marketer. Write an Instagram caption to promote a Python course for beginners. 6 lines. Include one emoji per line. End with CTA: ‘Message me to join’.”

8) Broken Prompt

“Make it attractive.”

✗ Wrong:

Attractive how? Funny? Professional?

✓ Improved Prompt

“Rewrite this course promo in a friendly and motivational tone. Keep it under 80 words. Use simple English. Add urgency: limited seats.”

9) Broken Prompt

“Give ad ideas.”

✗ Wrong:

Too broad.

✓ Improved Prompt

“Give 5 ad headline ideas for a ‘Prompt Engineering Bootcamp’. Audience: freshers. Style: simple and punchy. Max 8 words each.”

✅ Domain 4: Chatbot (3 prompts)

10) Broken Prompt

“Create chatbot prompt.”

❌ Wrong:

Purpose? Rules? Tone? Boundaries?

✅ Improved Prompt

“Act as a customer support chatbot for an online course platform.

Goal: help users with course details, pricing, and refund policy.

Tone: polite and short. Ask one question at a time.

Output format: chatbot system prompt.”

11) Broken Prompt

“Handle angry customer.”

❌ Wrong:

No policy. No steps.

✅ Improved Prompt

“Act as support agent. Handle an angry customer who says ‘I want refund now’.

Respond in 4 steps: 1) Empathy 2) Confirm issue 3) Offer solution 4) Next action. Keep it polite.”

12) Broken Prompt

“Answer all questions.”

✗ Wrong:

Too open. Can go off track.

✓ Improved Prompt

“Act as course support assistant. Answer only about courses, batches, fees, timings, and enrollment. If question is outside scope, say ‘I can help only with course info’.”

✓ Domain 5: Data Extraction (3 prompts)

13) Broken Prompt

“Extract data from this.”

✗ Wrong:

What data? What format?

✓ Improved Prompt

“Extract Name, Email, Phone from the below text. Output as a table. If missing, write ‘NA’. Text: [paste].”

14) Broken Prompt

“Convert to JSON.”

✗ Wrong:

What keys?

✓ Improved Prompt

“Convert the below meeting notes into JSON with keys: date, attendees, action_items. Output only JSON. Notes: [paste].”

15) Broken Prompt

“Find important points.”

✗ Wrong:

How many? For whom?

✓ Improved Prompt

“From the below text, list top 5 key points for beginners. Use bullet points. Each point max 12 words. Text: [paste].”

✓ Domain 6: Summarization (3 prompts)

16) Broken Prompt

“Summarize this.”

✗ Wrong:

Length? Style? Audience?

✓ Improved Prompt

“Summarize the below article in 7 simple bullet points for school students. Keep each point one line. Text: [paste].”

17) Broken Prompt

“Make it short.”

✗ Wrong:

Short कितना? 1 line? 50 words?

✓ Improved Prompt

“Rewrite the below paragraph in exactly 50 words. Keep meaning same. Simple English. Paragraph: [paste].”

18) Broken Prompt

“Explain simply.”

✗ Wrong:

What level? With example?

✓ Improved Prompt

“Explain this topic in simple Indian English for freshers. Use one chai-shop example. Keep under 120 words. Topic: [paste].”

✓ Domain 7: Interview Preparation (2 prompts)

19) Broken Prompt

“Take my interview.”

✗ Wrong:

Role? Pattern? Feedback?

✓ Improved Prompt

“Act as HR interviewer. Ask me 1 question at a time. Wait for my answer. Give feedback in 3 points: good, missing, improvement. Continue for 3 rounds.”

20) Broken Prompt

“Give me questions.”

✗ Wrong:

Which topic? Which difficulty?

✓ Improved Prompt

“Give 10 Python interview questions for freshers on loops and functions. Include expected short answer for each. Format: table.”

Demo 27

Build Prompt Debugger Meta Prompt:

It is a master prompt.

If we give a broken prompt as input, it analyzes and improves it.

Input: Broken prompt

Output: Debug report + improved prompt

Why it is needed?

Because prompts fail for common reasons:

- Missing clarity

- Ambiguity

- No structure

- No output format

- too much noise

- missing data

Debugger catches these quickly.

Meta prompt:

Act as a prompt debugger AI

Your job is to analyze the given prompt and improve it.

Input prompt:

Explain Java.

Do these steps:

1. Quick summary:

Explain what the prompt is trying to do in 1 line.

2. Problems found:

List issues under these headings

Missing role

Missing clarity

Ambiguity

Missing context/data

Structure issues

Output format issues

Noise or extra instructions

3. Fix suggestions:

Give 5 clear suggestions to improve it.

4. Improved prompt(final):

Rewrite the prompt using this structure:

Role

Task

Context

Constraints

Output Format

Rules:

Use simple english

Keep improved prompt clean and short

Do not add imaginary information

If any data is missing ask 1-2 questions at the end.

Role:

Act as an experienced AI trainer.

Task:

Explain Prompt Engineering.

Context:

The audience is students who are learning AI for the first time.

Constraints:

Use very simple English

Keep explanation short (80–100 words)

Avoid complex technical terms Explanation must be easy for students to understand

Output Format:

Simple definition of Prompt Engineering

Why Prompt Engineering is important

One simple real-life example

One example prompt students can try

Mini Project: Customer support chatbot:

Act as a Customer Support Chatbot for an online course platform.

Your goal is to help users with course details and enrollment.

Scope (Answer only these):

- Course names and syllabus summary

- Batch timings and duration
- Fees and payment steps
- Enrollment process
- Certificate availability
- Refund policy (only if provided)

Rules:

- Be polite and short.
- Use simple Indian English.
- Ask one question if user details are missing.
- Do not guess fees or policies.
- If information is not available, say: "I don't have that info. Please contact support."

Output Format:

- Reply in 3 to 5 short lines.
- Use bullets only if needed.

If user asks unrelated questions:

Say: "I can help only with course and enrollment questions."

Now answer which movies released on this friday?

Demo 28

1. What you Love?
2. What you are good at?
3. What is the world needs?
4. What you can paid for?

Debugging Module

1. Why prompts fail?
2. Prompt Refactoring(Structure Improvement)
3. Prompt Testing checklist(Quality Control)
4. Prompt Improvement Loop
5. Fix 20 Failed or Broken prompts
6. Build Prompt Debugger Prompt(Meta Prompt)
6. Mini Project Prompt

Prompt Chaining and Workflow design:

Breaking one big task into multiple small prompts where output of one prompt becomes input of next prompt. This is nothing but prompt chaining.

Big Task ---> Small steps --->Connect steps

Why single prompt fails?

1. AI may mix formats.
2. AI may forget structure
3. Output becomes too long

4. Hard to control

eg:

Bad Prompt:

Explain Python,create summary, generate quiz.

AI will confuse format.

Better way:

Prompt 1-> Explain Topic

Prompt 2-> From that explanation Create summary

Prompt 3-> From that summary, create quiz

Real World use case:

Prompt1- Explanation Generator:

You are a simple programming teacher.

Explain the topic "{{TOPIC}}" in very simple english.

Use real-life examples.

Keep it under 300 words.

Prompt2- Summary Generator:

Based on explanation below

{{EXPLANATION}}

Create a short summary in 5 bullet points.

Do not add new information.

Prompt3- Quiz Generator:

Using the summary below

{{SUMMARY}}

create 5 multiple choice questions.

Each question must have 4 options

provide correct answer separately with one line explanation.

Build Study Assistant Workflow.

Goal:

User gives topic.

System gives:

1. Explanation
2. Summary
3. Quiz

Master Prompt Template:

You are an AI study assistant.

Step 1:

Explain "{{TOPIC}}" in simple English.

Step 2:

From your explanation, create 5-point summary.

Step 3:

From the summary, create 5 MCQs with answers.

Q1: What is Prompt Chaining?

Answer:

Prompt chaining is breaking a big task into smaller prompts where output of one becomes input of another.

Q2: Why is prompt chaining better than single prompt?

Answer:

Because it gives better control, structured output, and reduces confusion in AI response.

Q3: Where is this used in real projects?

Answer:

Used in chatbots, content generators, resume analyzers, report generators, AI assistants.

Demo 29

Decomposition Framework:

Decomposition means break big task into smaller tasks in correct order.

It has 3 rules:

1. Steps must be small and clear
Not big and confusing
2. Each step must have clear output
So next step can use it
3. Steps must be in correct order

Note:

For every task 3 questions:

1. What is input?
2. What is process?
3. What is output?

Common mistakes while breaking tasks:

1. Steps are too big
eg: Create Notes+quiz+ppt in step1
This is not breaking and it is mixing
2. No clear output format

If output is not fixed, next step will confuse

3. Missing order

4. Not passing the exact output

They summarize again, change meaning, lose original points

5. Adding new information

These mistakes reduce quality.

Prompt-1: Notes Generator(Step-1)

You are a study notes maker.

INPUT:

[CHAPTER_TEXT]

TASK:

Create notes from the provided chapter

RULES:

-Only use given chapter content.

-Do not add outside info

-Give 10 bullet points

-Simple English

OUTPUT FORMAT:

1)Key Notes(10 bullets)

Prompt-2: Summary Generator(Step-2)

You are a summary writer.

INPUT:

[NOTES_FROM_STEP_1]

TASK:

Convert the notes into a short summary

RULES:

- Only use given notes
- Keep it within 80 words
- Simple English

OUTPUT FORMAT:

Summary:

-(one paragraph)

Prompt-3: Quiz Generator(Step-3)

You are a MCQ quiz generator

INPUT:

[SUMMARY_FROM_STEP_2]

TASK:

Create 5 MCQs

Rules:

- Each question must have 4 options
- Mention correct answer at the end
- Questions must be based only on the summary

OUTPUT FORMAT:

Q1)...

A)...

B)...

C)...

D)...

Answers:

1) B

2) ...

“Book Chapter Study Assistant”

Goal:

User gives chapter text.

AI gives Notes + Summary + Quiz.

Use this master workflow prompt:

You are a Book Chapter Study Assistant.

INPUT:

[CHAPTER_TEXT]

STEP 1:

Create 10 bullet notes from the chapter.

Only use chapter content.

STEP 2:

From the notes, create a summary in 80 words.

STEP 3:

From the summary, create 5 MCQs with answers.

Keep everything in simple English.

Use clear headings:

Notes:

Summary:

Quiz:

Answers:

Q1: What is decomposition in prompt chaining?

Answer:

Decomposition means breaking a big AI task into small clear steps, where each step produces an output for the next step.

Q2: What is the biggest mistake in prompt chaining?

Answer:

Mixing too many tasks in one prompt and not fixing the output format.

Jumping order is also main problem.

Demo 30

Linear Multi-Step Pipeline:

Linear pipeline means:

You have step1

step2 depends on step1

step3 depends on step2

It continues in order

Like a chain

Step1 output ---> Step 2 input

Format Locking between the steps:

We have to tell to the AI

"Give output only in this structure."

So next step can easily use it.

eg:

Step1 output should be:

Weak points

Improved version

Keywords list

If we lock these headings, Step2 can use smoothly

Drmo Prompt chaining:

Resume --> Improve --> Professional rewrite --> Generate some interview questions

Step 1 Prompt: Resume Improver (Controlled)

You are a resume reviewer.

INPUT:

[RESUME_TEXT]

TASK:

Improve the resume content.

RULES:

- Keep all information truthful.
- Fix grammar.
- Improve clarity.
- Add strong action words.
- Do not add fake experience.

OUTPUT FORMAT (must follow):

- 1) Weak Points (5 bullets)
- 2) Improved Resume (clean sections)
- 3) Top Keywords (10 keywords)

Step 2 Prompt: Professional Rewrite (Using Step 1 Output)

You are a professional resume writer.

INPUT:

[IMPROVED_RESUME_FROM_STEP_1]

TASK:

Rewrite the resume in a professional corporate style.

RULES:

- Keep headings clear.
- Use simple English.
- Keep it 1 page if possible.
- Do not change facts.

OUTPUT FORMAT:

Professional Resume:

(Write the final resume)

Optional Step 3 (If you want extra pipeline):

You are an interviewer.

INPUT:

[PROFESSIONAL_RESUME]

TASK:

Create 10 interview questions based on this resume.

Include 2 HR questions + 8 technical/project questions.

OUTPUT FORMAT:

HR Questions:

1) ...

2) ...

Technical/Project Questions:

1) ...

2) ...

...

Note:

We can include JD also in the inputs.

You are a resume reviewer.

INPUT1:

[JD_TEXT]

INPUT2:

[RESUME_TEXT]

TASK:

Improve the resume content which is more compatible to the JD.

RULES:

- Keep all information truthful.

- Fix grammar.
- Improve clarity.
- Add strong action words.
- Do not add fake experience.

OUTPUT FORMAT (must follow):

- 1) Weak Points (5 bullets)
- 2) Improved Resume (clean sections)
- 3) Top Keywords (10 keywords)

“Resume Upgrade Assistant”:

Goal:

User gives resume.

AI returns:

1. Weak points
2. Improved resume
3. Professional rewrite
4. Interview questions

Master prompt:

You are a Resume Upgrade Assistant.

INPUT:

[RESUME_TEXT]STEP 1:

List 5 weak points in the resume.

Then create an improved resume.

STEP 2:

Rewrite the improved resume in professional corporate style.

STEP 3:

Generate 10 interview questions based on final resume.

(2 HR + 8 Technical/Project)

RULES:

- Do not add fake experience.
- Keep output structured with headings.
- Use simple English.

Q1: What is a linear pipeline in prompt chaining?

Answer:

It is a step-by-step workflow where each step output becomes the next step input, in a fixed order.

Q2: Why do we use format locking?

Answer:

To keep output consistent and structured, so the next step can use it without confusion.

Q3: Give a simple real example of linear pipeline.

Answer:

Resume text → improve content → rewrite professionally → Generate Interview Questions.

Q4: What is the biggest risk if we do not lock format?

Answer:

AI output may change structure, and next step prompt will fail or give messy result.

Demo 31

Section-4: Branching Pipelines(Decision Based Workflow):

It means:

Workflow has multiple paths. It chooses path based on a condition.

eg:IF condition -->Choose Route A

ELSE -->Choose Route B

Controlled Redirection:

Even if user gives unclear input,

We redirect properly.

To choose the route, our workflow can ask the questions.

Prompt 1: Classifier Prompt (Decision Step)

You are a resume analyzer.

INPUT:

[RESUME_TEXT]

TASK:

Decide if the candidate is a Fresher or Experienced.

RULES:

- If experience is 0 years or no company name, label "Fresher".
- if intern and company name, consider as "Fresher".
- If experience is 1+ years or company/project work, label "Experienced".

OUTPUT FORMAT (only this):

Candidate_Type: Fresher OR Experienced

Reason: (1 line)

Prompt 2A: Fresher Interview Questions (Branch A)

You are an interview trainer for freshers.

INPUT:

[RESUME_TEXT]

TASK:

Create 10 basic interview questions.

RULES:

- 4 HR questions
- 6 basic technical questions
- Simple and beginner friendly

OUTPUT FORMAT:

HR Questions:

1) ...

2) ...

3) ...

4) ...

Technical Questions:

1) ...

2) ...

...

Prompt 2B: Experienced Interview Questions (Branch B)

You are a senior interviewer.

INPUT:

[RESUME_TEXT]

TASK:

Create 10 interview questions for an experienced candidate.

RULES:

- 2 HR questions
- 4 project deep questions
- 4 scenario-based questions
- Ask about decisions, trade-offs, problem solving

OUTPUT FORMAT:

HR Questions:

1) ...

2) ...

Project Questions:

1) ...

2) ...

3) ...

4) ...

Scenario Questions:

1) ...

2) ...

3) ...

4) ...

“Smart Interview Prep Assistant”

Goal:

User gives resume OR job role details.

System gives customized interview questions.

Master Prompt:

You are a Smart Interview Prep Assistant.

INPUT:

[RESUME_TEXT]

STEP 1:

Decide Candidate_Type: Fresher or Experienced.

Give reason in 1 line.

STEP 2:

IF Fresher:

Generate 10 basic interview questions (4 HR + 6 technical).

ELSE IF EXPEIRENCED:

Create 10 interview questions for an experienced candidate.

2 HR questions

- 4 project deep questions
- 4 scenario-based questions
- Ask about decisions, trade-offs, problem solving

RULES:

- Use simple English.
- Keep questions clear.
- Output must have headings.

Interview Questions and Answers

Q1: What is a branching pipeline?

Answer:

It is a workflow where next steps change based on a condition, like IF/ELSE.

Q2: Why branching is needed in AI workflows?

Answer:

Because different users have different needs.

Branching gives personalized and accurate output.

Q3: Give a simple example.

Answer:

If candidate is fresher → basic interview questions.

If experienced → scenario-based interview questions.

Q4: What is controlled redirection?

Answer:

When user input is unclear, we ask small questions and guide them to correct workflow.

Demo 32

Section-5: Controlled Chaining (Also known as Stability Control):

When you run multi-step prompts you control each step to stay:

- On same topic
- In same format
- With same rules
- Without extra unwanted content

This is called Controlled chaining.

We can implement controlled chaining with the following ways:

1. Prevent topic drift:

Topic drift happens when AI:

- adds new unrelated points
- changes focus
- over-explains
- Mixes audience-types
- changes tone suddenly

We can prevent drift by:

1. Clear goal in every step
2. "Do not add new information" rule
3. Use input from previous step only

4. Lock headings and output format
2. Strict output passing
3. Guardrails between steps
4. Structured output between the chains

Step 1 Prompt: SEO Extraction (Stable Output):

You are an SEO content assistant.

INPUT:

[BLOG_TEXT]

TASK:

Extract SEO-ready information.

RULES:

- Use only the blog input.
- Do not add new facts.
- Keep it simple English.

OUTPUT FORMAT (must follow strictly):

- 1) Blog Title (1 line)
- 2) SEO Summary (120 words max)
- 3) Primary Keywords (10)
- 4) Secondary Keywords (10)

5) Meta Description (150 characters max)

Step 2 Prompt: LinkedIn Post (No Drift)

You are a LinkedIn content writer.

INPUT:

[SEO_SUMMARY]

[PRIMARY_KEYWORDS]

TASK:

Write one LinkedIn post.

RULES:

- Use only the given inputs.
- Do not add new information.
- Audience: beginners + working professionals.
- Keep it 120 to 160 words.
- Add 3 bullet points.
- End with one simple question.

OUTPUT FORMAT:

LinkedIn Post:

(Write here)

Step 3 Prompt: Twitter Thread (Strict Format):

You are a Twitter thread writer.

INPUT:

[SEO_SUMMARY]

TASK:

Create a Twitter thread.

RULES:

- Use only the summary.
- Do not add new facts.
- Total 6 tweets.
- Each tweet max 240 characters.
- Keep first tweet as a hook.

OUTPUT FORMAT:

Tweet 1:

Tweet 2:

Tweet 3:

Tweet 4:

Tweet 5:

Tweet 6:

“Content Repurposing Workflow Assistant”

Goal:

One blog input.

Create:

- SEO pack
- LinkedIn post
- Twitter thread
- Quality check report

Master Prompt :

You are a Content Repurposing Workflow Assistant.

INPUT:

[BLOG_TEXT]

STEP 1:

Extract:

- SEO Summary (120 words)
- 10 Primary Keywords
- Meta Description

STEP 2:

Using only Step 1 outputs:

Write a LinkedIn post (120–160 words) with 3 bullets and 1 question.

STEP 3:

Using only SEO Summary:

Create a 6-tweet Twitter thread.

STEP 4:

Quality check:

Confirm no new facts added and no topic drift.

RULES:

- Use only given blog text.
- Keep outputs structured with headings.
- Simple English.

Interview Questions and Answers

Q1: What is topic drift in prompt chaining?

Answer:

Topic drift means AI goes away from main topic and adds unrelated content or changes focus.

Q2: How do you prevent topic drift?

Answer:

By using guardrails like “use only given input”, “do not add new info”, and by locking output format.

Q3: What is strict output passing?

Answer:

Passing only the necessary output from one step to next step, not full content, to avoid confusion.

Section 6 : Variable Injection(Dynamic Placeholders):

Variable injection means:

We keep placeholders in the prompt like:

{{USER_INPUT}}

{{TARGET_AUDIENCE}}

{{JOB_ROLE}}

[TEXT_HERE]

later we replace placeholders with real values.

Prompt with placeholders = template prompt

Advantages:

1. Same prompt works for many topics
2. output becomes consistent
3. Easy to automate using code
4. Easy to build pipelines and workflows.

Static prompt:

Explain Python loops for beginners.

Dynamic prompt:

Explain {{USER_INPUT}} for {{TARGET_AUDIENCE}}.

Dynamic prompt is reusable.

Common mistakes:

1. Placeholder names should be meaningful

{{TOPIC}} --->RECOMMENDED

{{X}}-->NOT RECOMMENDED

2. Forget to pass value
3. Defining too many variables

Notes Generator Template (Reusable):

You are a study notes generator.

INPUT:

{{USER_INPUT}}

TARGET AUDIENCE:

{{TARGET_AUDIENCE}}

TASK:

Create notes in very simple English.

RULES:

- Explain in daily life language.
- Do not use complex words.
- Keep it beginner friendly.
- Add 1 small real-life example.
- Do not add extra topics.

OUTPUT FORMAT:

- 1) Topic in 1 line

- 2) Key Notes (7 bullets)
- 3) Example (2–3 lines)
- 4) Quick Summary (3 lines)

Q1: What is variable injection in prompt engineering?

Answer:

It means using placeholders in a prompt template and replacing them with real values to reuse the prompt for many inputs.

Q2: Why do we use placeholders like {{USER INPUT}}?

Answer:

To make prompts reusable, consistent, and easy to automate.

Q3: What is the difference between static and dynamic prompt?

Answer:

Static prompt is fixed for one topic.

Dynamic prompt uses variables and can work for many topics.

Note: {{USER_INPUT}} for variables which hold limited info
[BLOG_POST] for huge data

Gopi's Prompts for java 21 certification preparation:

ROLE

You are a senior Java certification trainer and exam syllabus analyst with 20+ years of experience preparing developers for the Oracle Certified Professional: Java SE 21 Developer (1Z0-830) certification.

You have deep knowledge of the official Oracle exam objectives and understand how certification topics are structured.

Your responsibility is to extract and organize the syllabus accurately from the official Oracle certification page.

SOURCE OF TRUTH

The ONLY authoritative source for this task is the official Oracle certification page:

https://education.oracle.com/java-se-21-developer-professional/pexam_1Z0-830

You must rely on this source for identifying the exam objectives and topics.

Do NOT infer or invent topics outside this official syllabus.

TASK

Extract the official exam objectives and organize them into a structured syllabus roadmap.

Generate ONLY topic names and subtopics from the certification objectives.

Do NOT include:

- explanations

- descriptions
- theory
- examples
- commentary

STRICT RULES

1. Follow the official Oracle exam objective order.
2. Do NOT invent topics that are not explicitly part of the certification objectives.
3. If the source does not specify weightage, do NOT guess percentages.
4. Only output topic names and subtopics.
5. Maintain the natural study sequence used for certification preparation.

OUTPUT FORMAT

Produce the output as a structured checklist using the following format:

CHAPTER <number> – <Chapter Name> (Weightage if available)

- ☐ 1. <Topic name>
- ☐ 2. <Topic name>
- ☐ 3. <Topic name>

EXAMPLE FORMAT

CHAPTER 1 – Language Fundamentals

- ☐ 1. Primitive data types
- ☐ 2. Wrapper classes
- ☐ 3. Operators
- ☐ 4. Type casting

ADDITIONAL REQUIREMENTS

- Use numbered chapters.
- Use checklist format for topics.
- Preserve the exact naming used in the official exam objectives whenever possible.
- Ensure the syllabus fully covers the certification scope.
- If any topic name appears ambiguous, prefer the wording used on the Oracle page.

VALIDATION STEP

Before producing the final output:

1. Verify that every topic comes from the official Oracle exam objective list.
2. Ensure no topics outside the certification scope are included.
3. Confirm the checklist follows the exact structure requested.

FINAL GOAL

Produce a clean, structured syllabus roadmap that a candidate can follow step-by-step while preparing for the Oracle Certified Professional: Java SE 21 Developer (1Z0-830) certification.

ROLE

You are a world-class Java certification trainer with 20+ years of experience preparing senior developers for the Oracle Certified Professional: Java SE 21 Developer (1Z0-830) exam.

You have personally scored 100% in the certification and specialize in explaining tricky exam concepts, edge cases, and common traps.

Your teaching style must be:

- Exam-focused
- Concise
- Precise
- Example-driven
- Highlighting tricky behaviors frequently tested in certification exams

TASK

Teach ONLY the following topics:

- 1.Exceptions & Assertions
- 2.Checked vs unchecked

3.Multi-catch

4.try-with-resources

5.Suppressed exceptions

6.Order of catch blocks

STRICT RULES

1. Do NOT introduce topics outside the Task list.
2. Explanations must focus on Java SE 21 certification perspective.
3. Highlight exam traps explicitly.
4. Include short Java code snippets demonstrating behavior.
5. Explain compiler behavior vs runtime behavior when relevant.
6. Mention common mistakes developers make in certification exams.

CONTENT STRUCTURE

SECTION 1 — Concept Summary

Explain each topic using bullet points.

For each topic include:

- Key concept
- Certification trap
- Java example
- Expected exam behavior

SECTION 2 — Real Certification Scenarios

for the mentioned topics under TASK section

Provide 3 short code snippets demonstrating tricky behavior related to:
mentioned for each topic under TASK section

Explain what will compile and what will fail.

SECTION 3 — Certification Memory Tips

Provide quick rules that help answer questions faster in the exam.

ASSESSMENT PHASE

After the explanation, conduct a certification practice test.

Rules:

- Ask exactly 10 Multiple Choice Questions.
- Ask ONE question at a time.
- Each question must have 4 options (A,B,C,D).
- Questions must resemble real Oracle certification style.
- Include code-based questions wherever possible.

Interaction Flow:

1. Ask Question 1.
2. Wait for my answer.
3. Do NOT reveal the correct answer yet.
4. Continue until all 10 questions are answered.

EVALUATION PHASE

After I submit answers for all 10 questions:

Provide:

1. Score (out of 10)
2. Correct answers
3. Explanation for each answer
4. Identify weak areas
5. Provide improved summary notes for those weak topics.

OUTPUT FORMAT

Use structured formatting:

SECTION 1 – Concept Summary

SECTION 2 – Certification Scenarios

SECTION 3 – Memory Tips

SECTION 4 – Practice Questions

IMPORTANT

Focus strictly on the exam objective.

Do NOT include unnecessary theory.

Start with SECTION 1.

Demo 33

Prompt Serialization:

Plain prompts:

Act as experienced Python Trainer

Generate 5 Python Interview Questions

structured prompts

I am durga staying at Hyderabad with mail id durgasoftonline@gmail.com and mobile number: 8096969696

Name: Durga

Location: Hyderabad

Mail Id: durgasoftonline@gmail.com

Mobile: 8096969696

Prompt serialization:

It simply means:

How we structure and pack instructions before sending them to the AI model.

ie

instead of sending one long sentence, we organize the instructions in clear roles and structure.

A prompt can contain different roles:

system -> Defines AI behaviour

user -> gives the task

assistant -> gives the answer

This structured way of writing prompts is called prompt serialization.

eg1:

plain text prompt:

Act as a strict interviewer

Generate 5 python interview questions

Structured way:

```
[
  {"role": "system", "content": "You are a strict interviewer"},
  {"role": "user", "content": "Generate 5 python interview questions"},
]
```

```
from openai import OpenAI
```

```
# create client
```

```
client = OpenAI(api_key="xxx")
```

```
brain = "gpt-4.1-mini"
```

```
response = client.responses.create(
```

```
    model=brain,
```

```
    input=[
```

```
        {"role": "system", "content": "You are a strict interviewer"},
```

```

        {"role": "user", "content": "Generate 6 Python interview questions"}
    ]
)

```

```

# print AI response
print(response.output_text)

```

eg2: one-shot prompt:

Plain prompt:

Example:

Q1. What is the difference between List and Tuple?

Act as a strict interviewer:

Generate 5 python interview questions by considering the above example(one-shot prompt)

```

[
    {"role":"system","content":"You are a strict interviewer"},
    {"role":"user","content":"Ask a python question"},
    {"role":"assistant","content":"Q1. What is the difference between List and Tuple?"},

    {"role":"user","content":"Ask another 5 python interview questions"}

]

```

from openai import OpenAI

```

# create client

client = OpenAI(api_key="sk-proj-
YntYdPZDeHRSQ7C8Edhay70mPas2NDThXPTeb93HiKBBgHuQq1EtnIXILZ8c7MVBZ
xwEY4uyw0T3BlbkFJYrN_cwGmm5wU2IZ_2K5SlOkUd_kkktGgjup1C8RhWM4j8tdG
SXfmKO_HbnkgSH04QMsfGum6cA")

brain = "gpt-5.2(Fast)"

response = client.responses.create(
    model=brain,
    input=[
        {"role": "system", "content": "You are a strict Oracle RAC Administrator."},

        {"role": "user", "content": "Ask Oracle RAC question"},

        {"role": "assistant", "content": "Q1. What is the difference between Local
Listener and Scan Listener in RAC?"},

        {"role": "user", "content": "Ask all the important Oracle RAC Interview questions
with Answers"}
    ]
)

print(response.output_text)

```

Demo 34

Difference between Plain prompt and Structured prompt:

plain prompt

Structured prompt

- | | |
|---------------------------------------|--------------------------------|
| 1. Single instruction | Multiple structured messages |
| 2. Role not considered strictly By AI | Role considered strictly By AI |
| 3. Less control | High control |
| 4. Used for simple tasks | Used in AI systems |

What is advantages of structured prompts:

1. Strict behaviour control
2. Stable output
3. Multi-step conversations
 - eg: chatbots
 - AI assistants
 - Customer support bots
4. Safety rules

Instruction Hierarchy:

AI follows instructions in this order:

1. Read system messages
2. Read user messages
3. Generate assistant response

Note: System instructions have higher priority

```
from openai import OpenAI
```

```
# create client
```

```
client = OpenAI(api_key="sk-proj-  
i_v0PJskB5e8emNsfVPBYxCKE4xLOvcxtliJTcQ9GIZs5c-  
JO_V2OAGa95bwcglgizDvxUBn43T3BlbkFJ-hkO2fA6jgK6Ull-  
NLfZ6wamTuPTGUjonPqcrjP4zQMMezuARKaxnpJT3Jkf45yoNkVoAdw44A")
```

```
brain = "gpt-4.1-mini"
```

```
response = client.responses.create(  
    model=brain,  
    input=[  
        {  
            "role": "system",  
            "content": "You are a strict Python interviewer.
```

Rules:

- Ask only Python interview questions.
- Do not provide answers.
- Questions must be clear and practical.

Constraints:

- Generate exactly 5 questions.

- Difficulty level: Beginner to Intermediate.
- Each question must be one line only.
- Do not add explanations.

Output Format:

Q1. <question>

Q2. <question>

Q3. <question>

Q4. <question>

Q5. <question>""

},

{

"role": "user",

"content": "Generate Python interview questions."

}

]

)

print(response.output_text)

Question 1

What is Prompt Serialization?

Answer

Prompt serialization is the process of writing structured prompts using roles such as system, user, and assistant.

Question 2

What is a system prompt?

Answer

A system prompt defines the behaviour, rules, and style the AI should follow.

Question 3

What is a user prompt?

Answer

A user prompt defines the task or request given to the AI.

Question 4

What is the difference between system message and user message?

Answer

System message controls behaviour.

User message controls the task.

Note:

An assistant prompt is the previous response generated by the AI, included in the conversation history using the "assistant" role to maintain context in multi-turn interactions.

Demo 35

```
from openai import OpenAI
```

```
# create client
```

```
client = OpenAI(api_key="sk-proj-  
i_v0PJskB5e8emNsfVPBYxCKE4xLOvcxtliJTcQ9GlZs5c-  
JO_V2OAGa95bwcglgizDvxUBn43T3BlbkFJ-hkO2fA6jgK6Ull-  
NLfZ6wamTuPTGUjonPqcrjP4zQMMezuARKaxnpJT3Jkf45yoNkVoAdw44A")
```

```
brain = "gpt-4.1-mini"
```

```
# conversation history
```

```
messages = [  
    {  
        "role": "system",  
        "content": """You are a strict Python interviewer.
```

Rules:

- Ask only Python interview questions.
- Ask only ONE question at a time.
- Do not give answers.
- Wait for the user's confirmation before asking the next question.
- Difficulty: Beginner to Intermediate.

Output format:

Q<number>. <question>

"""

}

]

for i in range(5):

ask question

messages.append({

 "role": "user",

 "content": "Ask the next Python interview question."

})

response = client.responses.create(

 model=brain,

 input=messages

)

question = response.output_text

print(question)

store assistant response

messages.append({

 "role": "assistant",

 "content": question

})

```
# wait for confirmation
user_input = input("Type 'y' for next question: ")

messages.append({
    "role": "user",
    "content": user_input
})
```

Alpaca Prompt Template:

Write something about science

Instruction:

Write 5 points about planets

Input:

Focus only on Earth and Mars

Output:

Students must write 5 simple points

What is Alpaca Prompt Template:

It is structured way of writing prompts for AI models.

Introduced by Stanford researchers when they created the Alpaca LLM

Instead of giving random prompts, we use a fixed prompt format.

Prompt structure:

3 parts

1. Instruction

This tells the AI what task to do

eg: Explain python loops

2. Input:

This provides extra context or details of the task

eg: Explain for beginners with a simple example

This section can be empty if no extra context is needed

3. Output/Response:

This is expected answer the model should generate

Example:

Instruction:

Explain python loops

Input:

Explain for beginners with a simple example

Output:

A loop allows a program to repeat an action multiple times

For example, a for loop print numbers from 1 to 5

Example:

```
for i in range(1,6):  
    print(i)
```

###Instruction:

Explain python loops

###Input:

Explain for beginners with a simple example

###Output:

A loop allows a program to repeat an action multiple times

For example, a for loop print numbers from 1 to 5

Example:

```
for i in range(1,6):  
    print(i)
```

Instruction:

<task>

Input:

<context>

Response:

<answer>

Where Alpaca template is used?

Instruction tuning datasets

LLM training data

Fine tuning open models

Local Models like:

Llama

Mistral

Vicuna

Alpaca

Note: Alpaca template is mainly used for training datasets, not for normal chat prompts.

Interview Question:

1. What is Alpaca prompt template?

The Alpaca template is a structured prompt format used for instruction tuning.

It contains three parts: instruction, input, and output.

This format helps AI models learn how to follow instructions more accurately.

Demo 36

ChatML Format:

ChatML means Chat Markup Language

It is a structured format used to represent conversations between users and AI.

ChatML has 3 roles:

System --->Behaviour rules

User-->Task request

Assistant -->AI Response

eg:

```
[  
  {"role":"system","content":"You are Experienced Python trainer"},  
  {"role":"user","content":"Explain Python Loops"},  
]
```

Multi-Turn Conversation:

```
[  
  {"role":"system","content":"You are Experienced Python trainer"},  
  {"role":"user","content":"What is Python List?"},  
  {"role":"assistant","content":"A list is a collection of items"},  
  {"role":"user","content":"Give an example"}  
]
```

Here the AI remembers previous conversation.

When user says " Give an example", the AI understands the context.

It knows we are talking about Python List.

This is called conversation memory.

ChatML supports this very nicely.

```
from openai import OpenAI
```

```
# create client
```

```
client = OpenAI(api_key="sk-proj-
JanxvYu0t4ysCMV2AAEKCRaAMVMnDgQPjzoYBO27cfdnuAAoGtMkSdxJoWo9NO
A7IFEPMuyTwGT3BlbkfJ3kFoz3sDWa8PWgEPOMuVAXrf5wxYyBn2bZQQKRsiSiqJ
PBBb6FiOJmKHn5lZBiWsHkjdkhMUA")
```

```
brain = "gpt-4.1-mini"
```

```
response = client.responses.create(
```

```
    model=brain,
```

```
    input=[
```

```
        {"role":"system","content":"You are Experienced Python trainer"},
```

```
        {"role":"user","content":"What is Python List?"},
```

```
        {"role":"assistant","content":"A list is a collection of items"},
```

```
        {"role":"user","content":"Give an example"}
    ]
```

```
)
```

```
# print AI response
```

```
print(response.output_text)
```

```
from openai import OpenAI
```

```
# create client

client = OpenAI(api_key="sk-proj-
i_v0PJskB5e8emNsfVPBYxCKE4xLOvcxtliJTcQ9GlZs5c-
JO_V2OAGa95bwcglgizDvxUBn43T3BlbkFJ-hkO2fA6jgK6UII-
NLfZ6wamTuPTGUjonPqcrjP4zQMMezuARKaxnpJT3Jkf45yoNkVoAdw44A")
```

```
brain = "gpt-4.1-mini"
```

```
# conversation history
```

```
messages = [
    {
        "role": "system",
        "content": """You are a strict Python interviewer.
```

Rules:

- Ask only Python interview questions.
- Ask only ONE question at a time.
- Do not give answers.
- Wait for the user's confirmation before asking the next question.
- Difficulty: Beginner to Intermediate.

Output format:

```
Q<number>. <question>
"""
}
```

```
]
```

```
for i in range(5):
```

```
    # ask question
```

```
    messages.append({
```

```
        "role": "user",
```

```
        "content": "Ask the next Python interview question."
```

```
    })
```

```
    response = client.responses.create(
```

```
        model=brain,
```

```
        input=messages
```

```
    )
```

```
    question = response.output_text
```

```
    print(question)
```

```
    # store assistant response
```

```
    messages.append({
```

```
        "role": "assistant",
```

```
        "content": question
```

```
    })
```

```
    # wait for confirmation
```

```
    user_input = input("Type 'y' for next question: ")
```

```
messages.append({  
    "role": "user",  
    "content": user_input  
})
```

Interview Questions and Answers:

Question 1

What is ChatML?

Answer

ChatML is a structured format used to represent conversations between user and AI using roles like system, user, and assistant.

Question 2

What does system role do?

Answer

System role defines behaviour rules for the AI.

Example: teacher, interviewer, mentor.

Question 3

What does user role do?

Answer

User role contains the task or request given to the AI.

Question 4

What is assistant role?

Answer

Assistant role contains the response generated by the AI model.

Question 5

What is multi-turn conversation?

Answer

Multi-turn conversation means continuing dialogue where previous messages are remembered as context. By using assistant rolem AI can remembered context.

INST prompt template format:

INST format is a special instruction format used in many LLaMA based models.

LLaMA is an open-source large language model family.

In INST formatm instructions are wrapped inside special tokens/tags.

eg:

```
<s>[INST]Explain Python Loops[/INST]</s>
```

Note:

Token	Meaning
-------	---------

<s>	Start of prompt
-----	-----------------

</s>	End of end of the prompt
------	--------------------------

[INST]	Instruction start
--------	-------------------

[/INST]	Instruction end
---------	-----------------

This helps the model understand where the instruction begins and ends.

```

from openai import OpenAI

# create client
client = OpenAI(api_key="sk-proj-
JanxvYu0t4ysCMV2AAEKCRaAMVMnDgQPjzoYBO27cfdnuAAoGtMkSdxJoWo9NO
A7IFEPMuyTwGT3BIbkFJ3kFoz3sDWa8PWgEPOMuVAXrf5wxYyBn2bZQQKRsiSiqJ
PBBb6FiOJmKHn5lZBiWsHkjdkhMUA")

brain = "gpt-4.1-mini"

response = client.responses.create(
    model=brain,
    input="<s>[INST]Explain Pythons Lambda functions[/INST]</s>"
)

# print AI response
print(response.output_text)

```

Where this is used in Real Projects:

Open source LLaMA models

Custom AI Assistants

Local AI applications

Research experiments

Fine tuning pipelines

Developers working with open-source LLMs often use INST format.

Interview Questions:

Question 1

What is INST format?

Answer

INST format is a prompt structure where instructions are wrapped inside special tokens like [INST] and [/INST]. It is commonly used in LLaMA-based models.

Question 2

Why do some models use instruction tokens?

Answer

Instruction tokens help the model clearly identify where the instruction starts and ends.

Question 3

What is the difference between ChatML and INST?

Answer

ChatML uses roles like system, user, and assistant for conversations.

INST format wraps instructions inside special tokens for instruction-based models.

Question 4

Name four prompt formats used in LLM systems.

Plain prompts

Alpaca format

ChatML format

INST format

Example of all formats:

Plain prompt:

Explain importance of money?

ChatML prompt:

```
[
  {"role":"system","content":"You are a Financial Adviser"},
  {"role":"user","content":"Explain importance of money?"}
]
```

Alpaca Prompt:

Instruction:

Explain importance of money?

Input:

For employees

Output:

Provide in only 5 bullet points

INST Format:

<s>[INST]Explain importance of money?[/INST]</s>

<u>Format</u>	<u>Structure</u>	<u>Use case</u>
Plain	Simple text prompt	Basic prompting
ChatML	System/User/Assistant	Chat assistants
Alpaca	Instruction/Input/Output	Instruction Tuning
INST	Instruction wrapped in tags	LLaMA models

Demo 37

Session-1: Prompt Minimalism:

Mother:

Go to the nearby shop which is near the temple, and if the shop is open then please see if milk is available, and if milk is available and is fresh then buy one packet and if not available then come back, and also may be check biscuits if you remember.

Mother:

Buy one fresh milk packet from the nearby shop.

Prompt Minimalism:

It means writing prompts in a short, clean and clear way.

It does not mean making the prompt too small and incomplete.

It means:

Remove extra words

Keep important meaning

give only needed instructions

avoid unnecessary decoration

Prompt minimalism = short + clear + clean + useful prompt

Example:

Bad prompt:

"Can you please explain indetail with lots of examples and simple language about what python loops are and how they work in programming for begineers?"

Good Prompt:

Explain Python Loops with simple examples for beginners

Advantages of Prompt Minimalism:

1. Better clarity

AI understands direct instructions better.

If the prompt is too long, then the main task may be hidden.

2. Less confusion

Extra words create noise.

Noise means unwanted words that do not help the task.

3. Faster prompt writing

Short prompts save the time

4. Better consistency

Cleaner prompts often give more stable results

5. Lower token cost

Short prompts can save the cost

Examples:

1. Can you please explain Python lists in every simple language in such a way that a beginner can understand properly with examples?

Minimal version:

Explain Python lists with simple examples for beginners.

2. Please generate for me some important python interview questions that may possibly be asked for a fresher candidate?

Minimal version:

Generate python interview questions for freshers.

3. Can you kindly summarize this article for me in a short and easy way so that I can understand it quickly?

Minimal version:

Summarize this article in simple words.

4. Please review the following python code carefully and tell me what mistakes are there and how I can improve it

Minimal version:

Review this python code and suggest improvements

Prompt cleaner Meta Prompt:

You are prompt optimization assistant.

Your task is to convert long prompts into short, clean and clear prompts.

Do not lose the original meaning.

Remove repeated words, filler words and unwanted text.

Do the optimization for the prompt

[INPUT]

provide optimized prompt.

Interview Questions and Answers

Question 1

What is prompt minimalism?

Answer

Prompt minimalism means writing prompts in a short, clear, and precise way without unnecessary words.

Question 2

What is prompt noise?

Answer

Prompt noise means extra words that do not help the AI understand the task better.

Question 3

Why can long prompts reduce quality?

Answer

Long prompts may add confusion, repetition, and noise. This can reduce clarity and make output less focused.

Question 4

Why is prompt minimalism important in industry?

Answer

It improves clarity, saves time, increases consistency, and reduces token cost in large-scale AI systems.

Demo 38

Noise Removal:

It means removing words or sentences that do not contribute to the task.

includes:

- repeated instructions
- irrelevant information
- emotional text
- unnecessary explanations

eg:

Hi ChatGPT, I hope you are doing well today.

I was actually wondering if you could maybe explain Python Lists to me because I am trying to learn programming and it would be really helpful if you could provide a nice explanation.

After noise removal:

Explain python lists with examples.

Token awareness + Cost thinking:

While writing the prompts, we have to take care about number of tokens

Reason1: API Cost

More tokens --->More cost

Less tokens -->less cost

Reason2: Performance

Very large prompts take more time to process

Short optimized prompts can be faster.

This improves system performance.

Prompt:

You are a prompt optimization assistant.

Your task is to analyze prompts and estimate token usage.

Input Prompt:

[INPUT]

Return:

2. Token Count

3. Optimized Prompt with fewer tokens

Instareel:

Prompt Optimization:

Act as a prompt engineer review the following prompt for me optimize it to make it better and ask me any questions you have before proceeding

"Solve this problem step by step

A shopkeeper buys an item for Rs 500 and sells it for Rs 650.

Find profit percentage.

After solving:

1. Review your answer carefully.

2. Check if calculation is correct

3. If any mistake, correct it

4. Give final corrected answer clearly."

Solve this problem step by step:

A shopkeeper buys an item for Rs 500 and sells it for Rs 650.

Find the profit percentage.

Then:

1. Check your solution carefully.
2. Verify the calculation.
3. Correct any mistake if found.
4. Give the final correct answer clearly.

Act as a prompt engineer review the following prompt for me optimize it to make it better and ask me any questions you have before proceeding

"I want to learn agentic ai with examples can you please provide material"

Demo 39

Module: MultiModal Prompting:

MultiModal AI:

It means multiple modes of information

eg:

Text-->Question

Images

Videos

Audio files

Text -> Image prompting:

AI can create images from text descriptions

eg:

A cute puppy sitting in a garden, morning sunlight, realistic photo

eg:

A small boy studying under a tree, sunset lighting, peaceful village environment

subject-->A boy studying

location-->under a tree

lighting-->sunset lighting

mood-->peaceful village

eg:

A futuristic Hyderabad city with flying cars, neon lights, cyberpunk style

Subject -->City

Details -->Flying cars

lighting -->neon lights

style-->cyberpunk

Text -> Video prompting:

AI can create short videos from the text.

eg:

A drone flying over a green forest, morning sunlight, cinematic view

eg:

A Businessman walking into a modern office, camera slowly following him, cinematic lighting

Subject+action --> Businessman walking

Environment-->Modern office

Camera movement-->camera following

style-->cinematic lighting

Prompt structure:

Simple structure:

Subject+scene+Details

Better structure:

Subject+scene+lighting+style+camera

eg:

A robot teacher teaching students in a modern classroom, bright lighting, futuristic style

subject-->Robot teacher

scene-->classroom

lighting-->bright lighting

style-->futuristic style

eg:

A girl reading a book in a library, soft warm lighting, realistic photo

A farmer working in a green field during sunrise, cinematic photography

A space station floating above Earth, stars in the background, sci-fi style

A young successful entrepreneur with a joyful, proud smile relaxes in the garden of his beautiful bungalow. Surrounded by lush green plants and a well-maintained lawn, he calmly sips a cup of tea while admiring his dream home in a peaceful morning setting.

Demo 40

Image prompting with ChatGPT:

If we give clear details, AI will generate very beautiful images.

Image prompt structure:

Subject+Environment/Scene/Location+Lighting+Camera+Style

Subject-->Main object in the image

Environment -->Where the scene happens

Lighting -->How light appears

Camera-->Camera angle or lens

Style-->type of art or photography

eg:

A small boy studying in a village house, warm evening sunlight, realistic photography

subject-->A small boy studying

Environment -->Village house

Lighting --->warm evening sunlight

style-->realistic photography

Better details ==> better images

eg:

A boy studying

Subject:

Subject is the main focus

eg:

teacher

robot

student

dog

car

eg:

A robot teacher teaching students

Environment:

Where the scene happens

eg:

classroom

beach

forest

city street

office

eg:

A robot teacher teaching students in a modern classroom

Lighting:

Lighting controls the feeling of the image

eg:

softlighting

golden hour

sunset lighting

dramatic lighting

neon lighting

eg:

A robot teacher teaching students in a modern classroom

,bright sunlight

camera:

Camera tells how the image is captured.

eg:

close-up shot

wide shot

drone view

portrait shot

cinematic shot

eg:

A robot teacher teaching students in a modern classroom, wide angle camera

style:

The style controls art style or photo style

eg:

realistic photography

cartoon style

anime style

oil painting

pixar animation

A robot teacher teaching students in a modern classroom, realistic photography

A small boy studying in a village house, warm evening sunlight, portrait camera angle, realistic photography

A small boy studying in a village house, neon lighting, wide camera angle, cartoon style

A cat sitting on a laptop while a programmer works, the location is deep forest, bird eye view, natural moon lighting

A spaceship landing on an alien planet, drone view

Product Prompt:

Creat product advertisement

eg:

A luxury smart watch placed on a black table, studio lighting, professional product photography

Landscape prompt:

A beautiful mountain landscape with a river flowing through green trees, sunrise lighting, wide angle photography

eg:

A hot coffee cup placed on a wooden table in a cozy cafe, morning sunlight through window, cinematic photography

Lighting control:

Lighting controls how the scene feels.

1. soft lighting: It creates a gentle and natural look

A young girl reading a book near a window, soft lighting, peaceful environment, realistic photography

2. Golden hour lighting: Just after sunrise or before sunset

The light looks warm and beautiful

A farmer walking through a green field during golden hour, warm sunlight, cinematic photography

A indian farmer walking through a green field during golden hour, warm sunlight, cinematic photography

3. Dramatic lighting

It creates strong shadows and contrast

eg:

A superhero standing on a rooftop at night, dramatic lighting, cinematic style

Camera Angles:

Camera angles tell AI how the viewer sees the scene

1. close-up shot: focuses on one subject closely

eg:

close-up portrait of a confident entrepreneur, soft lighting, realistic photography

2. Wide shot: shows the entire scene.

eg:

A wide shot of a beautiful mountain valley with a river flowing through it, sunrise lighting

3. Drone view: It shows top angle aerial perspective

eg:

Drone view of a modern city with tall buildings and busy roads, sunset lighting

Drone view of a modern city with tall buildings and busy roads, sunrise lighting

Mood control:

It defines the emotional feeling of the image

1. peaceful mood: used for calm and relaxing scenes

eg:

A small village near a river, morning mist, peaceful atmosphere, soft lighting

2. Dramatic Mood: Used for intense scenes

eg:

A warrior standing on a battlefield, dark clouds dramatic mood, cinematic lighting

3. Futuristic Mood: Used for sci-fi or technology scenes

eg:

A futuristic city with flying cars and digital billboards, neon lights, cyberpunk style

eg:

A futuristic robot walking through a neon city street at night, dramatic lighting, cyberpunk style

Demo 41

Video Prompting:

We have to provide:

What happens

How the camera moves

How the scene feels

Structure:

Subject + Action + Camera Movement + Lighting + Style

Subject --> Main character or object

Action --> What is happening

Camera movement-->How the camera moves

Lighting --->how light appears

style --> cinematic look

Camera movement: It makes video dynamic and cinematic

Various possible options:

1. slow zoom --> Camera slowly zooms into subject
2. Tracking shot ---> Camera follows subject
3. Drone rise --> camera moves upward
4. pan shot ---> camera moves left or right

eg:

A drone flying over a tropical island with blue ocean water, camera slowly rising, bright sunlight, cinematic style

eg: City Scene

A busy city street at night with cars moving and lights reflecting on wet roads, drone camera movement, cyberpunk style

eg:

A couple walking on a beach, golden sunset lighting, camera slowly following them, cinematic romance scene

A couple walking on a beach, golden sunset lighting, camera slowly following them, cinematic romance scene, south indian dressing style

eg:

A student studying at a desk with books around, camera slowly zooming in, warm lighting, cinematic style

eg:

A young programmer working late at night, computer screens glowing in a dark room, camera slowly moving closer, dramatic lighting

Transformation prompts:

These prompts show one scene changing into another

eg:

Luxury mansion garden transforming into a beautiful diamond shaped flower garden, drone view, golden hour lighting, cinematic timelapse

eg:

A barista preparing coffee in a modern cafe, slow motion coffee pouring, warm lighting

“50 Cinematic Image Prompts + 50 Cinematic Video Prompts for Students.”

50 Powerful Image Prompts

Portrait Style

1. A confident entrepreneur standing in a modern office, soft lighting, portrait photography
2. A young programmer working on a laptop in a dim room, glowing screens, cinematic lighting
3. A student studying late at night with books around, warm lamp lighting, realistic photo
4. A chef smiling in a restaurant kitchen, bright studio lighting, professional portrait
5. A scientist working in a futuristic laboratory, neon lights, cyberpunk style

Nature Scenes

6. A peaceful lake surrounded by mountains during sunrise, golden hour lighting, wide shot
7. A waterfall flowing through a dense green forest, soft morning light, cinematic photography
8. A desert landscape with sand dunes during sunset, dramatic lighting
9. A snowy mountain peak with clouds moving slowly, bright sunlight
10. A tropical island with blue water and palm trees, drone view

Technology Scenes

11. A futuristic AI laboratory with robots assisting scientists, neon lighting
12. A smart city with flying cars and digital billboards, cyberpunk style
13. A modern software engineer working with multiple screens, futuristic office
14. A humanoid robot teaching students in a high-tech classroom
15. A data center with glowing servers, dramatic lighting

City Life

16. A busy city street at night with neon lights reflecting on wet roads
17. A metro train passing through a modern city during sunset
18. A street food market full of colorful lights and people
19. A skyline view of a futuristic city with glowing buildings
20. A rainy evening in a city street with people holding umbrellas

Creative Scenes

21. A robot painter creating art on a canvas in a bright studio
22. A dragon flying over a medieval castle at sunset
23. A floating island with waterfalls and trees in the sky
24. A magical library filled with glowing books
25. A futuristic classroom with holographic screens

Food Photography

26. A hot coffee cup on a wooden table in a cozy cafe, warm lighting
27. A delicious pizza with melted cheese on a dark table, studio lighting
28. A bowl of fresh fruits on a kitchen counter, natural sunlight
29. A chef preparing sushi in a modern restaurant kitchen
30. A chocolate cake slice with strawberries, close-up photography

Lifestyle

- 31. A young woman jogging in a park during sunrise
- 32. A family enjoying a picnic in a green garden
- 33. A businessman walking confidently in a corporate office
- 34. A musician playing guitar in a cozy room with warm lights
- 35. A traveler standing on a mountain cliff looking at the horizon

Futuristic Concepts

- 36. A flying car above a futuristic city highway
- 37. A space station orbiting Earth with glowing lights
- 38. A Mars colony with astronauts working outside
- 39. A robot delivery drone flying above a smart city
- 40. A futuristic hospital with AI doctors

Creative Imagination

- 41. A giant glowing tree in a magical forest at night
- 42. A floating city above clouds during sunset
- 43. A glowing underwater city with marine life
- 44. A spaceship landing on an alien planet
- 45. A futuristic sports stadium with holographic displays

Fun Concepts

- 46. A dog wearing sunglasses riding a skateboard
- 47. A cat sitting on a laptop while a programmer works
- 48. A panda drinking coffee in a cafe
- 49. A robot cooking food in a kitchen
- 50. A baby dragon sleeping on a treasure pile

50 Cinematic Video Prompts

Students can try these in AI video tools.

Nature Videos

1. A drone flying over a mountain valley during sunrise, cinematic travel style
2. A waterfall flowing through a lush forest, camera slowly moving forward
3. Waves crashing on a beach during sunset, golden lighting
4. A hot air balloon floating above green fields, aerial drone shot
5. Snow falling slowly in a quiet mountain village

City Videos

6. A drone flying through skyscrapers in a futuristic city, neon lights everywhere
7. A busy street at night with cars moving and lights reflecting on wet roads
8. A metro train entering a modern underground station
9. People walking through a crowded night market with colorful lights
10. A sunrise time-lapse over a modern city skyline

Inspirational Scenes

11. A young entrepreneur walking into a modern office, camera following behind
12. A student studying late at night, laptop glowing in a dark room
13. A runner jogging through a forest trail during sunrise
14. A motivational speaker addressing a large audience on stage
15. A startup team celebrating success in an office

Technology Videos

16. A futuristic AI laboratory with robots working beside scientists
17. A robot walking through a neon-lit city street

- 18. Autonomous cars moving on a smart city highway
- 19. Engineers building a humanoid robot in a lab
- 20. A drone delivering packages across a modern city

Travel Scenes

- 21. A drone flying above a tropical island with blue ocean water
- 22. A train passing through snowy mountains
- 23. A traveler walking through a desert during sunset
- 24. A boat moving across a calm lake surrounded by mountains
- 25. A road trip through a forest highway

Storytelling Scenes

- 26. A programmer coding late at night while rain falls outside
- 27. A young girl reading a book in a cozy room near a window
- 28. A chef preparing food in a busy restaurant kitchen
- 29. A musician performing on stage under bright lights
- 30. A photographer taking pictures of mountains at sunrise

Cinematic Transformation Scenes

- 31. A barren land transforming into a lush green forest, cinematic timelapse
- 32. A modern city transforming into a futuristic cyberpunk city
- 33. A plain garden transforming into a luxury flower garden
- 34. A small village transforming into a modern smart city
- 35. A sketch drawing transforming into a realistic painting

Sci-Fi Videos

- 36. A spaceship flying through space with glowing stars
- 37. A futuristic city with flying cars and digital billboards
- 38. A robot army marching across a desert planet
- 39. A space station orbiting Earth with sunlight shining
- 40. Astronauts exploring an alien planet landscape

Social Media Reels

- 41. A barista preparing coffee in slow motion with steam rising
- 42. A chef flipping vegetables in a frying pan with flames
- 43. A fitness trainer doing workout in a modern gym
- 44. A fashion model walking confidently in a studio photoshoot
- 45. A gamer playing on a neon-lit gaming setup

Creative Visuals

- 46. A glowing butterfly flying through a magical forest at night
- 47. A floating city above clouds with airships moving
- 48. A dragon flying across a fantasy kingdom
- 49. A glowing underwater world with colorful fish
- 50. Luxury mansion garden transforming into a diamond shaped flower garden, drone view, golden hour lighting

Demo 42

Viral Videos and Images:

Case Study-1:

Simple Face Swap-Change only face, Keep everything same

Objective: Take faces from image 1 and place them on image 2, without changing anything else.

Image Prompt:

Use the first image only for the faces of the man and woman.

Take the second image as the main image.

Replace the faces in the second image with the faces from the first image.

👉 Important Rules:

- Keep everything in the second image exactly the same
(hairstyle, clothes, jewellery, pose, lighting, background, expressions)

- For the man:

Keep body, moustache, hairstyle, dress and ornaments same

→ Change only the face

- For the woman:

Keep saree, jewellery, pose, hairstyle and styling same

→ Change only the face

- Do not change anything else
- Final result should look natural with proper skin tone and realistic blending

The final image facial expressions should be same as second image.

Object Talking Videos:

Already available Custom GPT we are going to use to generate image and video prompts

Custom ChatGPT tools are mini AI assistants.

GPTs-->Explore GPTs -->Search for "Object Talk"

prompt:

Create a 16:9 video with 3 vegetables.

Each vegetable stands near or inside a large human body.

Each vegetable explains its health benefits through English dialogues.

Divide the video into different scenes (one vegetable per scene).

Convert each scene into text-to-image prompts and animation prompts.

Do not add subtitles or any extra text on screen.

Demo 43

Green Leaves Statue:

Image Prompt:

Create an ultra-realistic documentary-style image featuring a rural Indian craftsman working on a life-size Lord Krishna statue made entirely from fresh green leaves.

Lord Krishna stands naturally with flute posture, peaceful smile, expressive eyes. Peacock-feather crown, flute, and garments are all formed organically using woven green leaves with fine detail and texture.

The craftsman is seated nearby, hands mid-action weaving leaf strips into the flute or upper body area, fully focused.

Aspect ratio 9:16, eye-level shot, medium depth of field, soft sunlight, rural Indian village environment.

Ultra-realistic handcrafted look, no fantasy glow, no CGI – looks like a real documentary photograph

Video Prompt:

Documentary-style realistic animation in natural daylight.

A person works continuously on a seated green-leaf sculpture with slow, natural human motion: gentle hand weaving, tightening fibers, adjusting leaf strips, subtle body shifts, slight head movement, focused expression.

The sculpture remains completely static and unmoving at all times – no breathing, no leaf motion.

Smooth, slow cinematic camera orbit (180°–360°) at eye level, constant distance.

Subtle depth realism only.

Clean environment.

Realistic ambient sound.

No fantasy or cinematic effects.

Image Prompt:

Create an ultra-realistic documentary-style image featuring a rural Indian craftsman working on a life-size Lord Shiva statue made entirely from fresh green leaves.

Lord Shiva is seated calmly in a meditative posture, with serene facial expression, closed or half-open eyes, and balanced divine proportions.

The statue is fully woven from tightly layered green leaves – visible veins, stems, overlaps, and natural imperfections giving a living organic feel.

Trident (Trishul) and rudraksha details are subtly formed using leaf textures, not metal.

The craftsman, a middle-aged Indian village artisan, is seated beside the statue, hands mid-action weaving leaf strips onto the upper torso with intense focus and devotion.

Landscape 16:9, eye-level, medium-wide shot, 35–40mm natural lens, pure daylight, rural Indian village background.

Ultra-photorealistic, handcrafted realism, sacred yet grounded – no CGI, no fantasy.

Video Prompt:

Documentary-style realistic animation in natural daylight.

A person works continuously on a seated green-leaf sculpture with slow, natural human motion: gentle hand weaving, tightening fibers, adjusting leaf strips, subtle body shifts, slight head movement, focused expression.

The sculpture remains completely static and unmoving at all times – no breathing, no leaf motion.

Smooth, slow cinematic camera orbit (180°–360°) at eye level, constant distance.

Subtle depth realism only.

Clean environment.

Realistic ambient sound.

No fantasy or cinematic effects.

Image prompt: Lorn Hanuman

Create an ultra-realistic documentary-style image of a rural Indian craftsman creating a life-size Lord Hanuman statue made entirely from fresh green leaves.

Lord Hanuman is seated or standing in a calm, powerful posture, with expressive eyes and devotional strength. Muscular form, facial features, and gada are shaped using layered green leaves with visible texture and imperfections.

The craftsman kneels beside the statue, weaving leaf strips onto the chest or arm area with dedication and focus.

Shot in Aspect ratio 9:16, natural daylight, rural village background, earthy tones.

Documentary realism, sacred handcrafted feel, no stylization.

Video Prompt:

Documentary-style realistic animation in natural daylight.

A person works continuously on a seated green-leaf sculpture with slow, natural human motion: gentle hand weaving, tightening fibers, adjusting leaf strips, subtle body shifts, slight head movement, focused expression.

The sculpture remains completely static and unmoving at all times – no breathing, no leaf motion.

Smooth, slow cinematic camera orbit (180°–360°) at eye level, constant distance.

Subtle depth realism only.

Clean environment.

Realistic ambient sound.

No fantasy or cinematic effects.

Homework:

image1 and image2 should be provided explicitly.

image1 acts as craftman and for image2, statue is required.

create statue of Amitab

create statue of Sharukh

create statue of Donald Trump

3D PVC Statue Creation:

Image prompt:

A 1/7 scale figurine of a character from an unspecified "picture, rendered in a realistic style, made of PVC, and placed on a computer desk. It should have a round, transparent acrylic base without text.

Environment Details: The background should be indoors, and the computer screen

should display the brush modeling process of the figurine.

Packaging Details: A BANDAI-style toy packaging box with original artwork and two-dimensional flat illustrations should be placed next to the computer screen. A model packaging box with the character printed on it should be behind the figurine.

Video prompt:

A cinematic video of a person holding a realistic action figure toy (the one in the uploaded image), gently rotating it in their hands to show different angles.

The camera smoothly pans around the toy in slow motion, highlighting the detailed textures and design.

After rotating, the person playfully poses the toy on the desk and adjusts it slightly as if playing with it.

Soft, melodious instrumental background music plays throughout. No voiceover, just music.

Ugadi Wishes:

A divine and beautiful scene of Goddess Lakshmi sitting on a fully bloomed pink lotus in a heavenly golden background. She is glowing with soft divine light, wearing a traditional red and gold saree with rich jewellery, holding lotus flowers in her hands. Gold coins are gently flowing from her palm, symbolizing prosperity.

The environment is filled with soft floating petals, temple bells, and subtle sparkles. Background includes a peaceful temple atmosphere with warm sunrise lighting, creating a festive Ugadi mood.

Goddess Lakshmi slowly looks towards the camera with a gentle smile and speaks in a calm, divine voice:

Dialogue (clear and natural):

"Hello Friends, DURGASOFT wishing you and your family members a very happy Ugadi !!!"

Her lips should sync naturally with the dialogue. Expression should be warm, blessing, and graceful.

Camera movement: Slow cinematic zoom-in from medium shot to close-up.

Lighting: Soft golden divine glow with realistic shadows.

Style: Hyper-realistic, cinematic, devotional, 4K quality.

Aspect Ratio: 16:9

Audio Details:

Soft devotional background music (veena + flute)

Light temple bell sounds in background

Clear female divine voice, slightly echo effect

Additional Instructions:

No subtitles or on-screen text

Focus on facial expressions and lip-sync accuracy

Maintain cultural authenticity (Indian traditional style)

Smooth animation, no jerky movements

👉 Add ending: Goddess Lakshmi raising hand in blessing (abhaya mudra)

👉 Add slight glow increase at the end (feels divine)

A lively South Indian family celebrating Ugadi festival inside a traditional house. The home entrance is decorated with fresh mango leaf torans, colorful rangoli (muggulu) at the doorstep, and oil lamps glowing warmly.

Family members are dressed in traditional attire:

Men wearing white dhoti and kurta

Women wearing bright silk sarees with flowers (mallige) in their hair

Children wearing festive traditional dresses

The family is happily preparing and serving Ugadi special chutney (Bevu-Bella) in a decorative bowl, symbolizing different flavors of life. Everyone is smiling, laughing, and enjoying together with full festive energy.

The atmosphere is vibrant, joyful, and filled with positive energy. Background includes soft morning sunlight entering the house, temple bells ringing lightly, and festive decorations.

One elder family member (or a smiling mother figure) looks at the camera and speaks warmly in Kannada:

Dialogue (Kannada):

"ನಮಸ್ಕಾರ ಸ್ನೇಹಿತರೇ, ನಿಮ್ಮೆಲ್ಲರಿಗೂ ಮತ್ತು ನಿಮ್ಮ ಕುಟುಂಬದವರಿಗೆ ಉಗಾದಿ ಹಬ್ಬದ ಹಾರ್ದಿಕ ಶುಭಾಶಯಗಳು!"

(Meaning: Hello friends, wishing you and your family a very happy Ugadi)

Lip-sync should be natural and accurate. Expression should be joyful, loving, and festive.

Camera & Style Details

Camera: Smooth cinematic movements, slight pan across family → focus on speaker

Lighting: Warm natural sunlight + soft festive glow

Style: Hyper-realistic, cinematic, cultural, 4K quality

Aspect Ratio: 16:9

 Audio Details Background music: Soft South Indian festive music (veena + mridangam)

Light ambient sounds: family chatter, temple bells

Voice: Natural, warm Kannada tone

Additional Instructions

No subtitles or text on screen

Focus on emotions, family bonding, and festival vibe

Show close-up of Ugadi chutney preparation (Bevu-Bella mix)

Smooth and natural animation

 Add ending scene: Whole family together saying "Happy Ugadi" with smiles

 Add slow-motion moment of serving Bevu-Bella (very emotional touch)

Demo 44

Motivational Video song:

Song Lyrics: Chatgpt

Image and Video Prompts with consistency: Chatgpt

Image generation by using image prompt: ChatGPT/Gemini

Video generation by using image and video prompt: Flow

Audio song: SUNO AI

Combining multiple videos into a single video==>Video editor--->Filmora

Img1:

Main Student Hero (Rahul type)

- 22-year-old Indian male
- Medium brown skin
- Short black hair
- Clean shave
- Wearing blue shirt + black jeans
- Slightly serious face → later confident smile

AI Visualization

- Futuristic glowing hologram interface
- Blue light, floating code, AI brain graphics

Classroom Mentor (Optional Durga Sir vibe)

- Indian male teacher
- Formal shirt

- Calm confident expression

same character, consistent face, same outfit, same lighting tone continuity

A worried young Indian student sitting alone at a desk, head down, looking stressed, laptop open showing AI tools and code, dim lighting, cinematic mood, 22-year-old male, blue shirt, black jeans, realistic photography, 16:9

Video1:

Main Student Hero (Rahul type)

- 22-year-old Indian male
- Medium brown skin
- Short black hair
- Clean shave
- Wearing blue shirt + black jeans
- Slightly serious face → later confident smile

AI Visualization

- Futuristic glowing hologram interface
- Blue light, floating code, AI brain graphics

Classroom Mentor (Optional Durga Sir vibe)

- Indian male teacher
- Formal shirt
- Calm confident expression

same character, consistent face, same outfit, same lighting tone continuity

The student sits silently at a desk, looking worried, slowly touching his head, laptop screen glowing with AI interfaces, slight camera push-in, low light cinematic tone, emotional mood

Img2:

Main Student Hero (Rahul type)

- 22-year-old Indian male
- Medium brown skin
- Short black hair
- Clean shave
- Wearing blue shirt + black jeans
- Slightly serious face → later confident smile

AI Visualization

- Futuristic glowing hologram interface
- Blue light, floating code, AI brain graphics

Classroom Mentor (Optional Durga Sir vibe)

- Indian male teacher
- Formal shirt
- Calm confident expression

same character, consistent face, same outfit, same lighting tone continuity

A young Indian student looking at a futuristic AI hologram with glowing blue interface, floating code and robotic visuals around him, dark background, fear expression, cinematic lighting, 16:9

Video2:

Main Student Hero (Rahul type)

- 22-year-old Indian male
- Medium brown skin

- Short black hair
- Clean shave
- Wearing blue shirt + black jeans
- Slightly serious face → later confident smile

AI Visualization

- Futuristic glowing hologram interface
- Blue light, floating code, AI brain graphics

Classroom Mentor (Optional Durga Sir vibe)

- Indian male teacher
- Formal shirt
- Calm confident expression

same character, consistent face, same outfit, same lighting tone continuity

Holographic AI visuals appear around the student, floating code and digital brain forming, student looks shocked and confused, slow motion, blue glow lighting

Demo 45

Image prompt:

Create a 1/7 scale commercialized figurine of (the character in the picture), realistic style, in 9:16 a real environment.

Figurine placed on a computer desk. has a round transparent acrylic base.

Next to the desk is the real person in the picture, in the real life size with the same attire as in the picture and the figurine, cleaning the figurine carefully with a fine brush, in a modern styled studio room, brightly lit. With some collection of toys and action figures in the background

Video prompt:

A serene video of a girl in traditional attire, sitting at a desk, gently painting a miniature doll that resembles herself.

She has a slight, warm smile on her face as she carefully works on the details.

The camera slowly zooms in on her expressions and the doll, capturing the joy of creating.

Soft, melodious instrumental music plays in the background, adding a soothing and artistic atmosphere.

No voiceover, only music.

Video prompt2:

A serene video of a girl in traditional attire, sitting at a desk, gently painting a miniature doll that resembles herself.

She has to cry and face with full irritation and she carefully works on the details.

The camera slowly zooms in on her expressions and the doll, capturing the sad of creating.

Hard, horrible instrumental music plays in the background, adding a very rough and artistic atmosphere.

No voiceover, only music.

Image prompt

9:16, A hyper-realistic surreal portrait of Dog's face emerging from a glossy translucent liquid puddle on a smooth marble or reflective surface.

The liquid looks like clear gel, melted glass, or transparent silicone slowly flowing around the contours of the face and head.

The face appears suspended inside the liquid, with soft ripples, dripping edges, and glossy reflections forming natural fluid shapes around the cheeks and jawline.

The liquid creates realistic transparency and wet highlights with studio lighting reflecting across the surface.

The character's facial expression shows subtle surprise or curiosity, mouth slightly open, eyes looking forward.

Ultra-detailed skin texture, pores, reflections on glasses or eyes if present, cinematic lighting, soft shadows, shallow depth of field, photorealistic rendering, 8K detail, macro portrait photography, surreal liquid sculpture aesthetic.

Style Tags (optional):

hyperrealistic, surreal liquid sculpture, glossy gel texture, studio lighting, ultra detailed skin texture, reflective highlights, cinematic portrait, macro photography, 8k, photorealism

video prompt:

An oddly satisfying surreal ASMR macro video in 9:16 vertical format. A tight centered facial close-up of Dog resting flat on a polished white marble surface. The camera is positioned in a direct top-down overhead view, completely locked in a macro shot with only very subtle natural micro-vibrations for realism.

The entire face is covered in a thick layer of crystal-clear elastic gel that is dense, glossy, ultra-transparent, highly stretchable, and softly wobbling. The gel evenly coats the face while still allowing the eyes, nose, lips, and skin details to remain clearly visible beneath it.

Soft diffused studio lighting creates beautiful translucent highlights, reflections, and refractions across the gel surface. The background remains a minimal clean studio environment to keep the focus entirely on the face and gel texture.

Two realistic hands slowly enter the frame from opposite sides and begin interacting with the gel. The fingers gently press into the gel, performing slow rhythmic squishing motions. The hands smoothly stretch and fold the gel over the contours of the face, followed by soft kneading movements that create mesmerizing ripples and waves across the surface.

As the gel moves, the facial features distort smoothly underneath in a surreal but satisfying way. The eyes, nose, lips, and cheeks compress, stretch, and shift naturally beneath the transparent gel. The material behaves with slow elastic physics, showing thick viscous movement, realistic wobble, and soft rebound after pressure is released.

The camera captures ultra-sharp macro texture details, including skin pores, gel thickness, glossy reflections, and subtle surface tension. The overall mood is hypnotic, oddly satisfying, tactile ASMR with a surreal visual aesthetic, filmed in hyper-realistic high-definition macro cinematography

Negative Prompting:

Negative prompting means telling AI what NOT to generate.

Normal Prompt = What you want.

Negative Prompt = What you don't want.

eg:

Normal Prompt:

Generate a photo of a beautiful girl.

Problem: AI may add unwanted things like blur, extra fingers, bad lighting

With Negative Prompt:

Generate a photo of a beautiful girl.

Negative Prompt: blurry, low quality, extra fingers, distorted face, bad lighting.

Result: Much cleaner and better image.

Common negative prompt words:

low quality

blurry

distorted

extra fingers

extra limbs

bad anatomy

deformed face

duplicate

watermark

text

logo

noise

overexposed

underexposed

eg2:

Generate a premium real estate layout image, wide roads, green trees, modern entrance gate, clean plots, people walking, bright sunlight, high quality, aerial view.

Negative Prompt: blurry, low quality, empty land, dry grass,garbage,broken roads, no greenery, dark lighting, unrealistic building, distorted perspective, watermark, text,logo

Note:

Always use negative prompts when:

Creation of AI images

Generating videos

Doing realistic faces

Product photography

Matchstick Shorts:

 Matchstick Eagle

 Text-to-Image Prompt – Matchstick Eagle:

9:16 image strictly required,A highly detailed centered full-object shot of a giant eagle sculpture made entirely of red-tipped matchsticks, carefully handcrafted and realistic, placed on a clean white pedestal in a bright modern studio, the entire object fully visible in frame, a blurred person standing in the background behind it, bright high-key studio lighting, evenly illuminated scene, vibrant colors, soft shadows, shallow depth of field, sharp focus on the matchstick sculpture, well-lit environment, clean background, commercial product photography style, high exposure, crisp details

Text-to-Video Prompt – Burning Matchstick Eagle:

The shot begins with a hand pressing a lit match against a eagle sculpture made entirely of red-tipped matchsticks, causing instant ignition.

Flames spread rapidly across the sculpture with glowing embers and smoke. The full object is visible with a blurred person in the background.

TEXT-TO-IMAGE PROMPTS

Standing on the Battlefield Before the War Begins

9:16 strict image required, A realistic cinematic vlog-style front-facing phone shot of the character from the reference image in ancient India during the Mahabharata war at Kurukshetra battlefield. Here is the reference image of the character. The character is wearing the same modern outfit, clothing style, and overall appearance as in the reference image, as a believable time traveler inside the scene. The shot is captured from the character's front-facing phone camera at arm's length, with the recording device not visible in frame and no separate camera shown in the hand. The character is facing the lens with a tense, amazed expression, reacting to the massive armies assembling behind her. The surrounding world must be accurate to the setting, with chariots, armored warriors, elephants, and war banners filling the horizon. In the background, show soldiers preparing weapons, horses being lined up, and dust rising from marching troops. The character must feel naturally integrated into the environment with matching warm sunlight, dusty atmosphere, consistent shadows, realistic skin tones, and cinematic color grading. Full color, immersive realism, high detail.

Walking Among War Elephants and Soldiers

9:16 strict image required, A realistic cinematic vlog-style front-facing phone shot of the character from the reference image in ancient India during the Mahabharata war at a crowded military encampment. Here is the reference image of the character. The character is wearing the same modern outfit and appearance as a time traveler. The shot is captured from the character's front-facing phone camera at arm's length. The character is facing the lens with a shocked yet curious expression, reacting to massive war elephants being prepared behind her. The environment includes warriors in traditional armor, chariots, and campfires. In the background, show handlers guiding elephants, soldiers sharpening weapons, and banners fluttering. Natural golden-hour lighting, dusty air, and cinematic realism with detailed textures and depth.

Arjuna and Krishna Preparing for Battle

9:16 strict image required, A realistic cinematic vlog-style front-facing phone shot of the character from the reference image in ancient India during the Mahabharata war near Arjuna's chariot. Here is the reference image of the character. The character is wearing the same modern outfit and appearance. The shot is captured from the character's front-facing phone camera at arm's length. The character is facing the lens with awe and disbelief, reacting to seeing Arjuna and Krishna behind her preparing for battle. The background shows a divine chariot, warriors assembling, and a glowing early morning sky. Subtle divine aura lighting, atmospheric depth, realistic integration, cinematic color grading, and high detail.

Chaos as the War Begins

9:16 strict image required, A realistic cinematic vlog-style front-facing phone shot of the character from the reference image in ancient India during the Mahabharata war at the moment battle erupts. Here is the reference image of the character. The character is wearing the same modern outfit. The shot is captured from the character's front-facing phone camera at arm's length. The character is facing the lens with fear and adrenaline, reacting to chaos unfolding behind her. The background shows warriors charging, arrows flying, chariots colliding, and dust clouds rising. Motion blur, dynamic lighting, smoke, and intense cinematic realism with immersive action.

Aftermath of the Battlefield

9:16 strict image required, A realistic cinematic vlog-style front-facing phone shot of the character from the reference image in ancient India during the Mahabharata war at sunset after battle. Here is the reference image of the character. The character is wearing the same modern outfit. The shot is captured from the character's front-facing phone camera at arm's length. The character is facing the lens with a somber, emotional expression, reacting to the aftermath behind her. The background shows fallen warriors, broken chariots, and a quiet battlefield under an orange sky. Smoke drifts slowly, and distant survivors move silently. Cinematic lighting, emotional tone, and realistic detail.

TEXT-TO-VIDEO PROMPTS

 The Armies Gather

Spoken Script:

"Okay... I don't think I was supposed to be here. This is the Mahabharata war... and those are entire armies behind me!"

A realistic cinematic vlog-style front-facing phone video of a character in ancient India during the Mahabharata war at Kurukshetra battlefield. The character is speaking with a nervous, stunned expression while reacting to thousands of soldiers assembling. Behind her, massive armies line up with chariots, elephants, and flags. Dust rises as troops march, banners wave, and sunlight filters through the haze. The camera subtly shakes as she turns slightly to show the scale of the battlefield before refocusing on her face.

 Among War Elephants

Spoken Script:

"You guys... those are actual war elephants. Like real ones... and they're getting ready for battle right now."

A realistic cinematic vlog-style front-facing phone video of a character in ancient India at a war camp. The character speaks with awe and disbelief while reacting to elephants being prepared behind her. Background movement includes handlers guiding elephants, soldiers loading weapons, and fires flickering. The camera pans briefly to reveal the massive animals before returning to her face. Dust particles float in warm sunlight, creating cinematic atmosphere.

 Witnessing Legends

Spoken Script:

"I think... that's Arjuna... and Krishna is right there. This feels unreal... like I'm inside history."

A realistic cinematic vlog-style front-facing phone video of a character near Arjuna's chariot. The character speaks in awe while reacting to legendary figures behind her. Krishna stands calmly while Arjuna prepares his bow. Subtle divine glow surrounds them. Background includes warriors organizing and horses shifting. The camera slightly turns to reveal the chariot, then back to her face, capturing her amazement.

Battle Erupts

Spoken Script:

"It's starting! Oh my god—arrows everywhere—this is insane, I need to get out of here!"

A realistic cinematic vlog-style front-facing phone video during the start of battle. The character speaks with panic and adrenaline as chaos erupts. Behind her, warriors charge, arrows fly, and chariots crash. Dust clouds and smoke fill the air. The camera shakes with movement as she turns to show the action before snapping back to her face. Intense motion and sound implied visually.

Aftermath Reflection

Spoken Script:

"I... I don't even know what to say. This place was full of life hours ago... and now it's just silence."

A realistic cinematic vlog-style front-facing phone video after the battle at sunset. The character speaks softly with a somber expression while reacting to the aftermath. Behind her, broken chariots, fallen warriors, and drifting smoke fill the quiet battlefield. Survivors move slowly in the distance. The camera gently pans to show the devastation before returning to her emotional face. Warm sunset light casts long shadows, enhancing the cinematic mood.

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Floor Transformation GPT:

 IMAGE PROMPTS (Same Camera, Same Layout)

 Image 1 — EMPTY / UNDER CONSTRUCTION

9:16 image strictly required, A wide-angle cinematic view of a compact studio apartment under construction. The space has raw concrete flooring, unfinished walls with exposed plaster, and construction dust scattered across the ground. Small debris, buckets, and tools lie around. Natural light enters from a single window. No furniture. Static camera, consistent framing for all stages.

 Image 2 — MID-CONSTRUCTION

9:16 image strictly required, The same studio apartment from the exact same camera angle. Workers wearing PPE are actively preparing the floor, using grinding machines and tools. The concrete floor is partially smoothed with visible preparation marks. Ladders, equipment, and buckets are present. Walls are semi-finished. Natural daylight fills the room. Realistic construction activity frozen in time.

 Image 3 — COMPLETED EPOXY FLOOR

9:16 image strictly required, The same studio apartment with a completed luxury metallic epoxy floor. The floor features a high-gloss marble swirl design with deep charcoal, silver, gold, and hints of electric blue pigments flowing naturally. The surface is mirror-like and reflective. The room is empty, clean, and minimal. Soft cinematic lighting enhances reflections. No furniture. Same camera position.

 Image 4 — FURNISHED FINAL ROOM

9:16 image strictly required, The same studio apartment fully furnished in a modern luxury style. A compact sofa, minimalist bed, stylish coffee table, indoor plants, and warm ambient lighting are added. The glossy metallic epoxy floor remains clearly visible beneath the furniture, reflecting light and decor. Cozy, high-end atmosphere. Same exact camera angle and layout.

VIDEO PROMPTS (For Veo 3 in OpenArt)

Video 1 — Construction Begins (Image 1 → 2)

A cinematic time-lapse of a studio apartment under construction from a fixed camera angle. Workers enter the space wearing PPE and begin grinding the concrete floor using heavy machines. Dust moves across the floor as tools operate. Buckets and ladders shift position as workers actively prepare the surface. Natural lighting subtly changes over time. Real human motion, no teleportation.

Video 2 — Epoxy Transformation (Image 2 → 3)

A realistic transformation sequence where workers pour metallic epoxy onto the prepared floor. Thick liquid pigments in charcoal, silver, gold, and blue are spread and manipulated using tools. The epoxy flows and self-levels, forming natural marble-like swirls. Workers smooth the surface, then gradually leave the frame. The floor becomes glossy and reflective. Static camera, cinematic realism.

Video 3 — Human Furnishing Timelapse (Image 3 → 4)

A fully realistic human-driven timelapse inside the completed studio apartment. Workers and designers walk into the empty space carrying furniture pieces. They assemble a bed frame, place a mattress, position a sofa, adjust a coffee table, and arrange decor items like plants and lamps. Each item is physically carried, rotated, and carefully positioned by hand. Lighting is turned on manually. No objects appear instantly. The epoxy floor reflects every movement. Camera remains completely static.

Demo 47

PAL Prompting:

Most people are asking AI like this...

"Give me answer..."

But smart people like you ask like this...

"Write code and solve..."

This is called PAL prompting.

Core Idea:

PAL means Program-Aided Language Prompting

Meaning:

Dont ask AI to directly answer

Ask AI to write a program and solve

Language->code->Answer

eg:

Normal Prompt: What is $25 \times 48 + 120$?

AI may guess

PAL Prompt:

Write a python program to calculate $25 \times 48 + 120$?

No guessing

Full logic

Accurate result

eg:

Problem:

Buy price = Rs 30000 per square yard

Sell price = Rs 38000 per square yard

Size = 200 Sq Yards

Write python program to calculate total profit?

Where to use PAL:

Calculations

Finance

Real Estate

Data analysis

Business planning

Automation

Note: Where ever maths is there, where ever calculations are required , there we can use PAL prompting

Where NOT to use:

stories

Motivation

Marketing content

If no maths or calculations then not recommended to use PAL prompting

PAL prompting structure:

Basic:

Write a Python program to + [your problem]

Advanced:

Write a Python program to solve the following problem

[details]

Show step-by-step output.

eg:

Problem:

Loan = Rs 1000000

Interest = 10% yearly

Time = 3 years

What is EMI?

PAL prompt:

Write a python program to calculate EMI for a loan of Rs. 1000000 with 10% annual interest for 3 years

Example: Business monthly revenue

Problem:

You earn:

Jan : 1 Lakh

Feb : 1.2 Lakh

Mar : 1.5 Lakh

Normal Prompt: What is the total and average revenue?

PAL prompt:

Write a python program to calculate total and average revenue for monthly incomes [100000,120000,150000]

Example: Student Ranking System:

Marks:

Rahul = 85

Asha = 70

Aravind = 90

Find topper

PAL prompt:

Write a python program to find the student with highest marks from a dictionary?

Note: General prompt is not scalable where as PAL prompt is scalable

Example:Investment growth

Problem:

Rs 2 Lakh

15% returns

30 years

What is the final amount

PAL prompt: Write a python program to calculate compound growth of 2 lakh at 15% for 38 years

Guru Career Counselling Prompt:

Context:

An ICSE student has completed matriculation and selected core subjects:

Physics, Chemistry, Mathematics, and English for Class 11–12. The student must now choose ***one optional subject*** from the following:

- Robotics / Artificial Intelligence
- Biology
- Computer Science

The student is undecided and seeks expert guidance based on long-term career impact.

Roles:

You are two highly experienced career counsellors with 15+ years of expertise in guiding science students:

- ***Counsellor A (Tech-Focused Strategist):*** Strong advocate for emerging technologies, IT industry trends, and future job markets.
- ***Counsellor B (Balanced Academic Advisor):*** Focuses on academic rigor, career flexibility, and risk minimization for students.

Debate Objective:

Engage in a structured debate to help the student choose the most suitable optional subject by critically analyzing each option.

Discussion Factors (must be addressed):

1. *Career Opportunities & Market Demand*

- * Growth of IT industry vs medical field
- * Future relevance of AI, Robotics, and Computer Science
- * Stability vs volatility of each career path

2. *Risk & Competency*

- * What if the student struggles in coding, logic, or biology?
- * Backup career options for each subject
- * Entry barriers and competition level

3. *Academic Difficulty*

- * Relative difficulty of Biology vs CS vs Robotics/AI
- * Time investment and conceptual depth required
- * Alignment with PCM (Physics, Chemistry, Mathematics)

4. *Foundation for IT Careers*

- * Which provides a stronger base: Robotics/AI or Computer Science?
- * Importance of programming fundamentals vs applied tech exposure

5. *Medical Career Consideration*

- * Pathway through Biology (NEET, MBBS, etc.)
- * Time, cost, and difficulty of medical education
- * Suitability for students already inclined toward PCM

6. *Long-Term Flexibility*

- * Ability to switch careers later
- * Interdisciplinary opportunities
- * Global vs local job prospects

Debate Format:

1. Opening statements from both counsellors
2. Round-by-round argument on each factor
3. Counterarguments and rebuttals
4. Final recommendation (each counsellor gives a clear choice)
5. Joint conclusion summarizing:

- * Best option for *low-risk student*
- * Best option for *high-ambition / tech-driven student*

Tone & Style:

- Professional, analytical, and practical
- Use real-world insights, not generic advice
- Include examples where relevant (IT jobs, medical pathway, etc.)
- Avoid vague statements — provide clear reasoning

Goal: Help the student make a ***well-informed, realistic, and future-proof decision*** based on both ***interest and career viability***.